

Linear Transformations

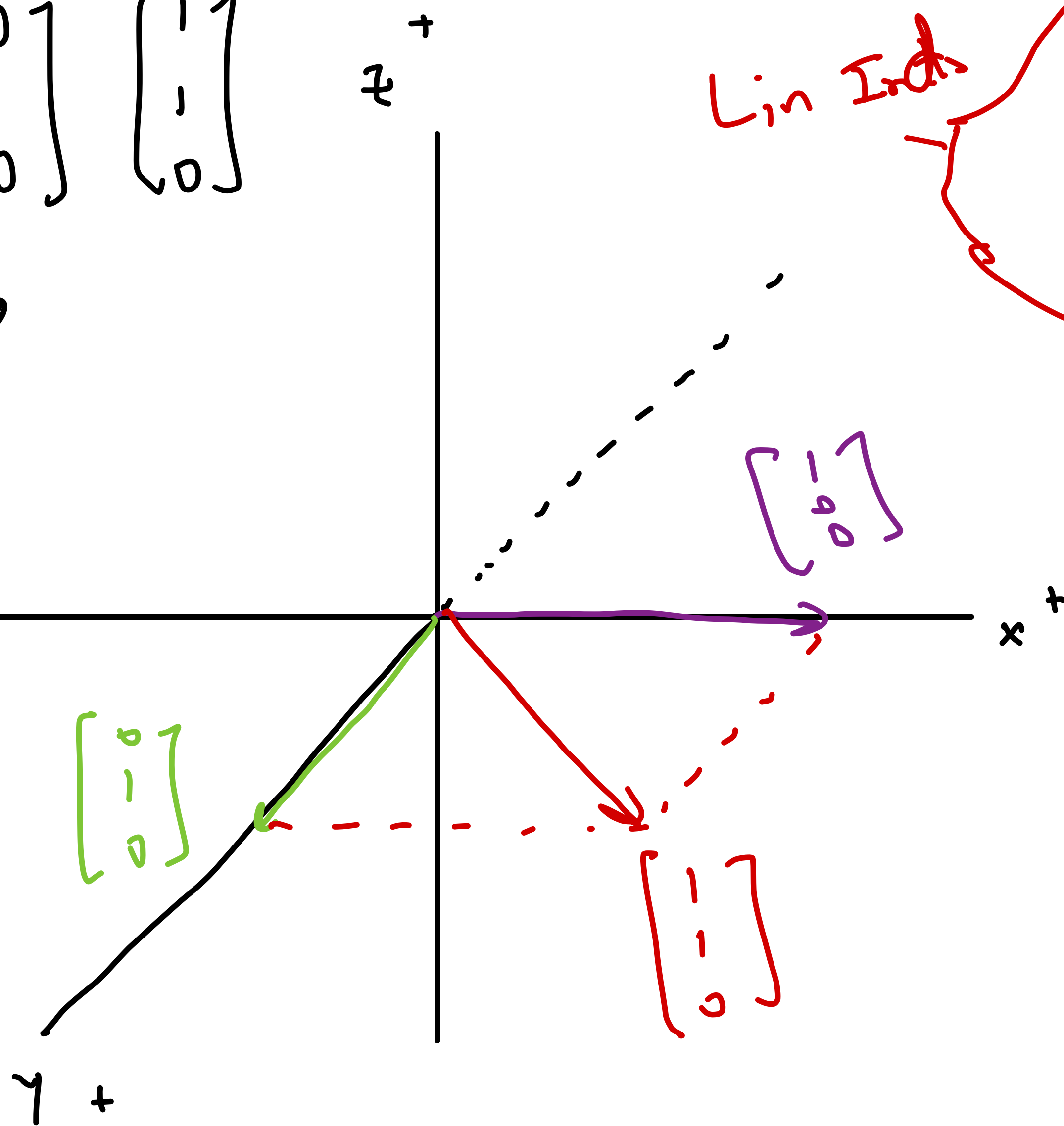
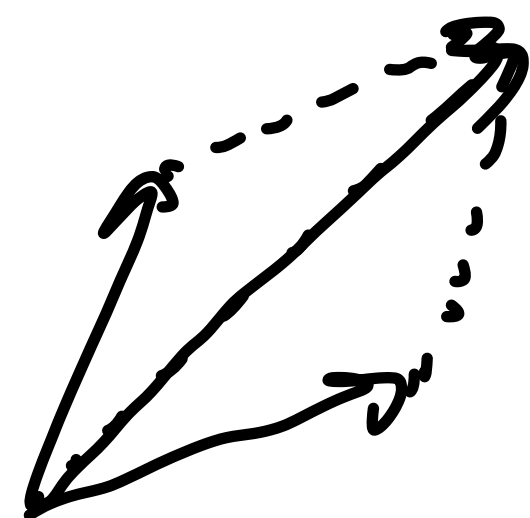
Geometric Algorithms
Lecture 7

Practice Problem

Find three vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ in $\mathbb{R}^3$ such that

- » every pair of vectors (i.e., $\{\mathbf{v}_1, \mathbf{v}_2\}$, $\{\mathbf{v}_1, \mathbf{v}_3\}$, $\{\mathbf{v}_2, \mathbf{v}_3\}$) is linearly independent
- » $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly dependent

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$



Lin Ind.

Lin. Ind

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 3 \end{bmatrix}$$

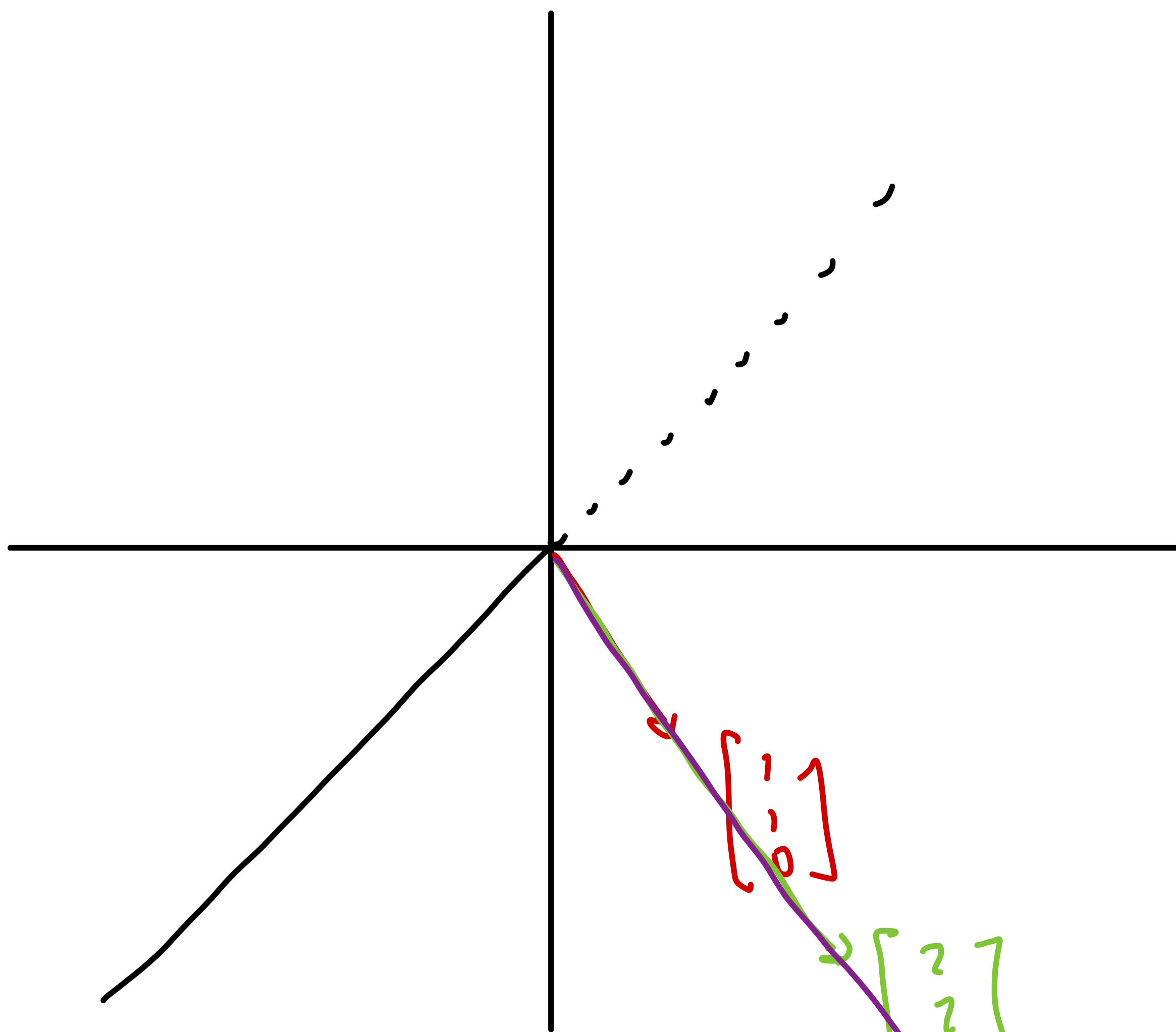
$\vec{v}_1 + \vec{v}_2 = \vec{v}_3$

Lin. Ind.

Lin. Dep.

$$\vec{v}_1 + \vec{v}_2 - \vec{v}_3 = 0$$

Dep. rel.



$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 3 \\ 0 \end{bmatrix} \right\}$$

Lin. Dep

$$\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 3 \\ 3 \\ 0 \end{bmatrix}$$

Objectives

- » Introduce matrix transformations
- » Define linear transformations
- » Start looking at the geometry of linear transformations

Keywords

Transformations

Domain, Codomain

Image, Range

Matrix Transformations

Linear Transformations

Additivity, Homogeneity

Dilation, Contraction, Shearing, Rotation

Recap

Recap: Homogenous Linear Systems

Definition. A linear system is *homogeneous* if it can be expressed as

$$A\mathbf{x} = \mathbf{0}$$

Recap: Linear Independence

Definition. A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is *linearly independent* if the vectors equation

$$x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \dots + x_n\mathbf{v}_n = \mathbf{0}$$

has exactly one solution (the trivial solution)

Recap: Linear Dependence

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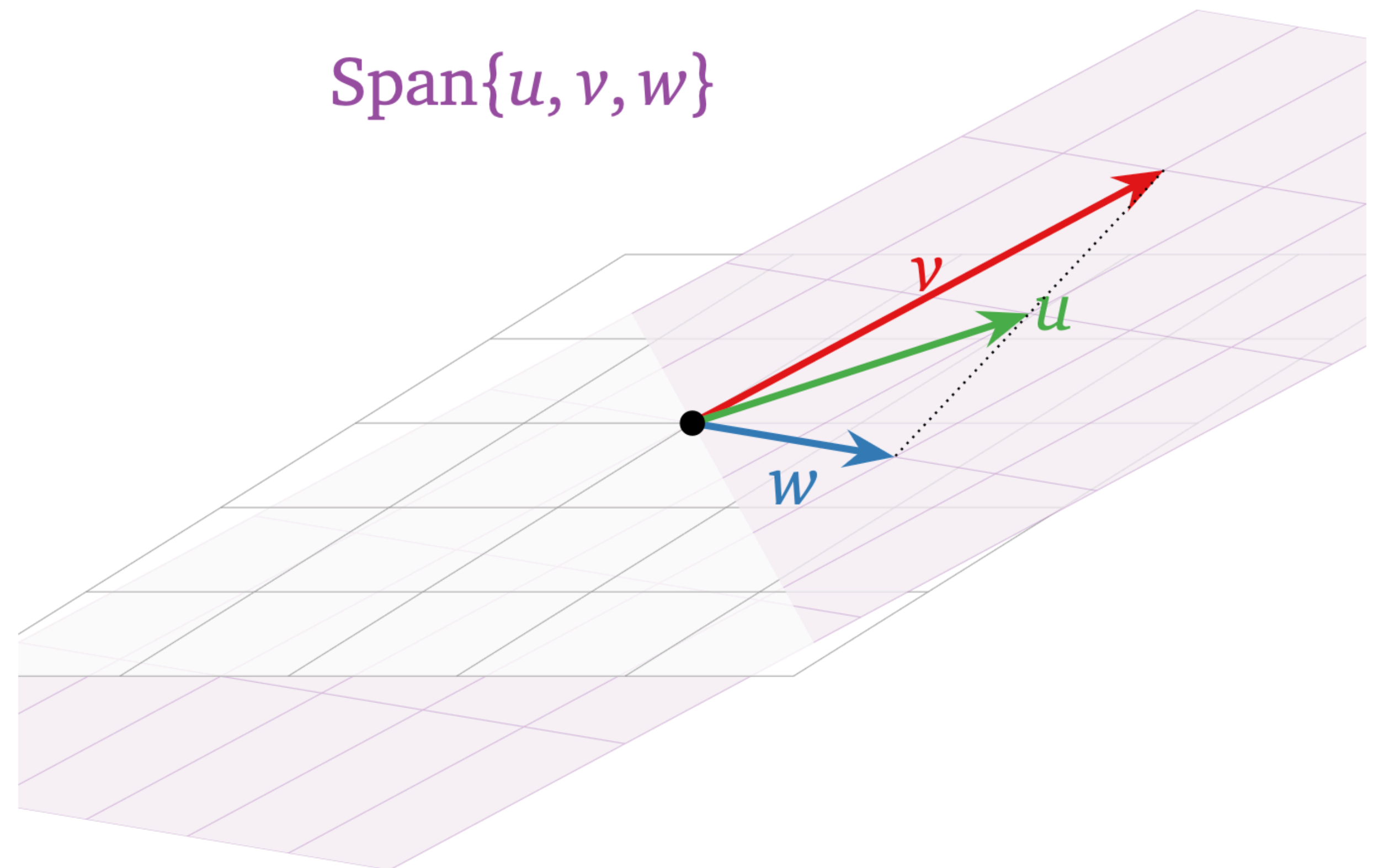
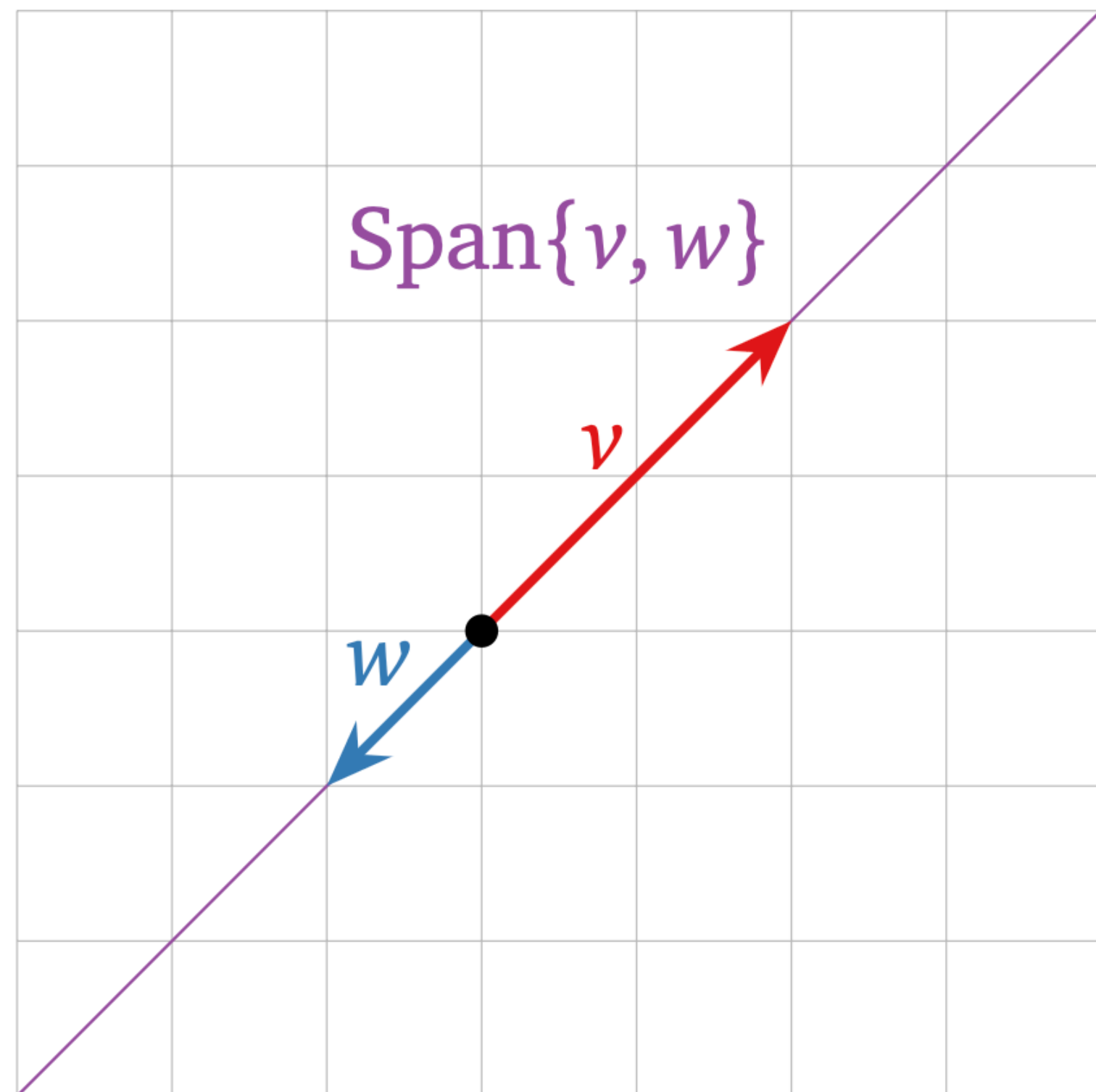
$$x_1 \mathbf{v}_1 + x_2 \mathbf{v}_2 + \dots + x_n \mathbf{v}_n = \mathbf{0}$$

has a *nontrivial* solution

Recap: Linear Dependence

Definition. A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is *linearly dependent* if one of its vectors can be written as a linear combination of the others (not including itself)

Linear Dependence (Pictorally)



Recall: Linear Dependence Relation

Definition. If $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are linearly dependent, then a ***linear dependence relation*** is an equation of the form

$$\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_n \mathbf{v}_n = \mathbf{0}$$

A linear dependence relation
witnesses the linear dependence

Recap: Increasing Span

Theorem. $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are linearly
dependent if and only there is an
 $i \leq n$,

$$\mathbf{v}_i \in \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{i-1}\}$$

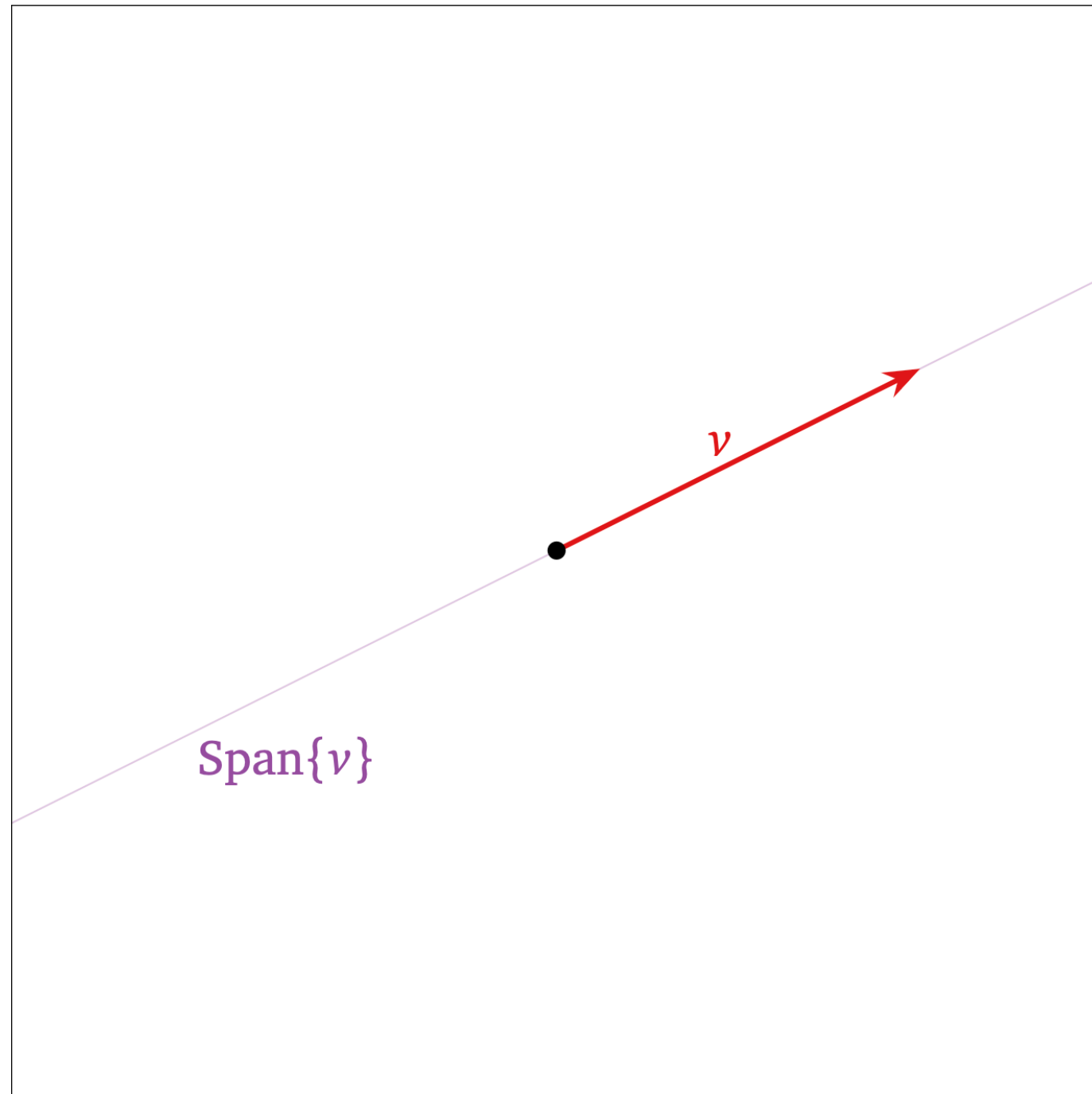
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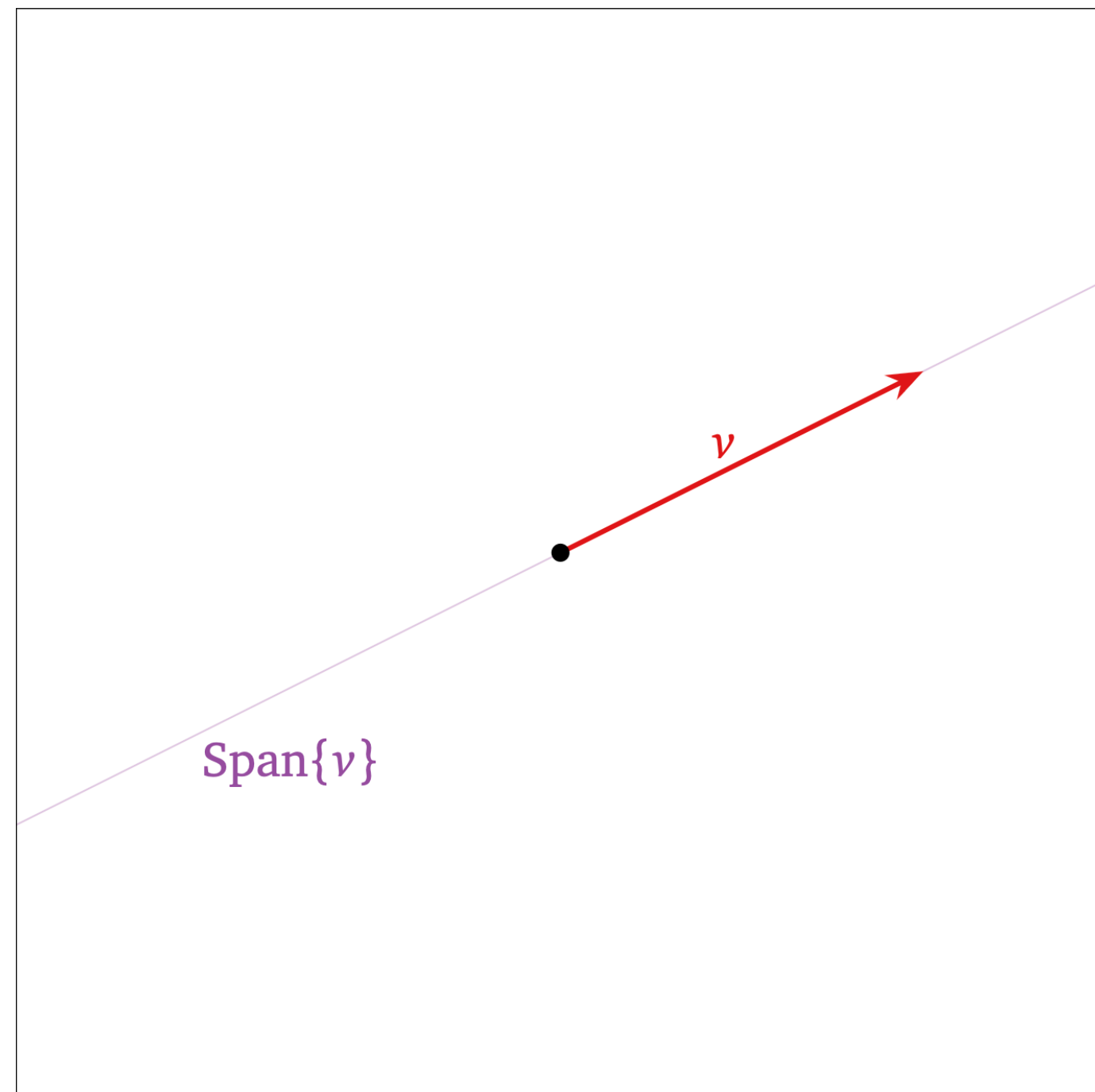
As we add vectors, we'll eventually find
one in the span of the preceding ones

Recap: Increasing Span

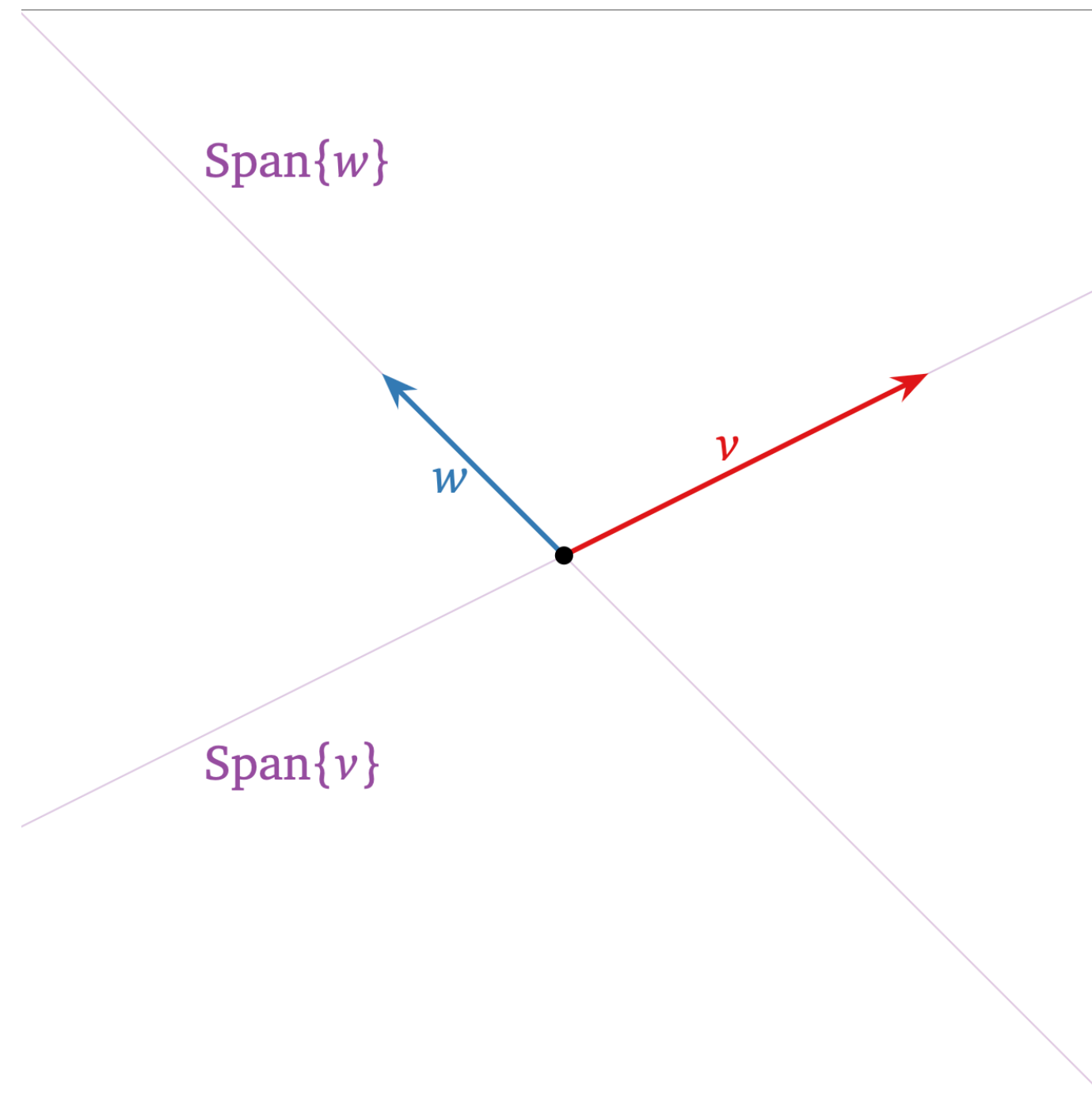


span of 1 vector
a line

Recap: Increasing Span

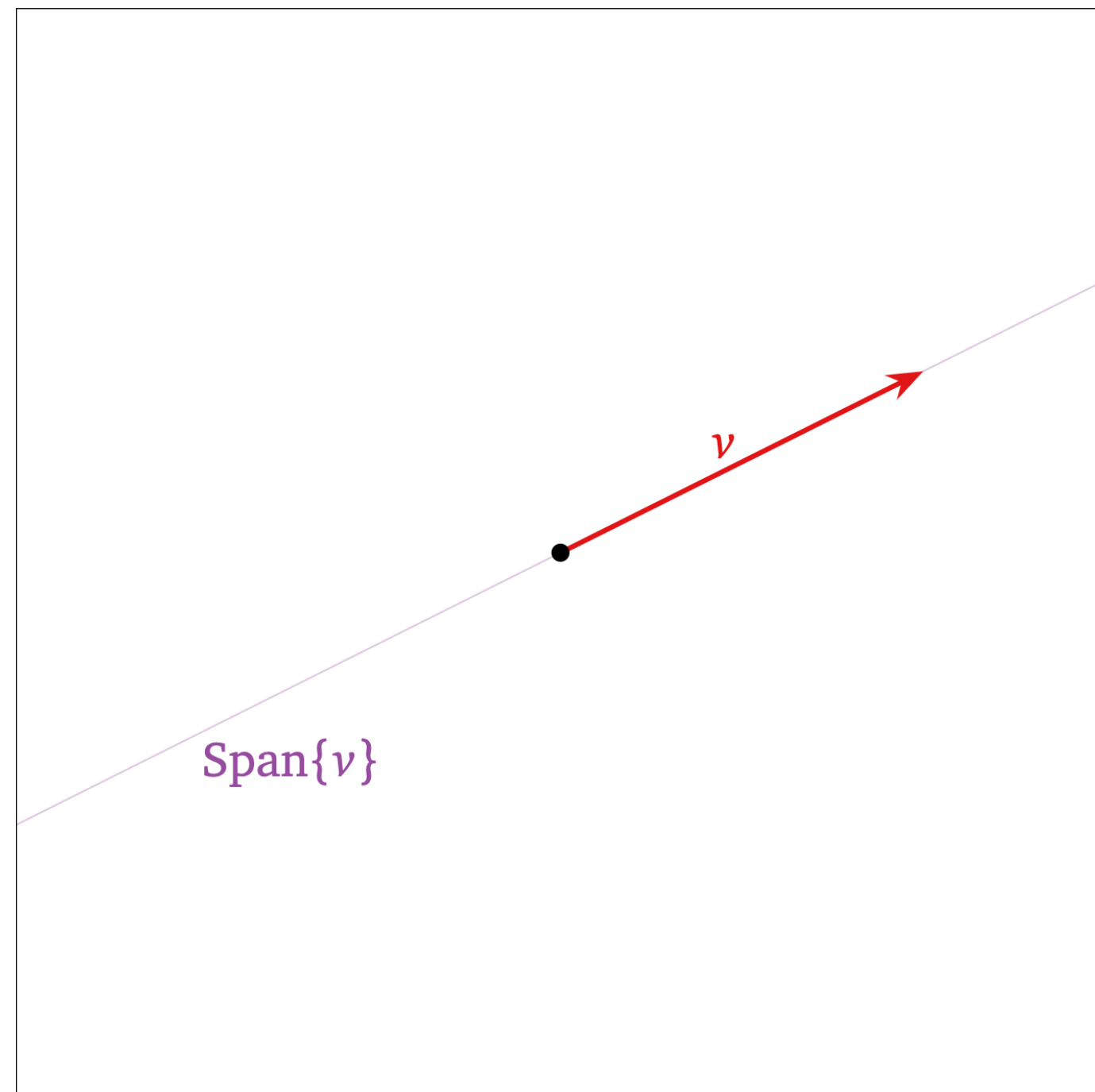


span of 1 vector
a line

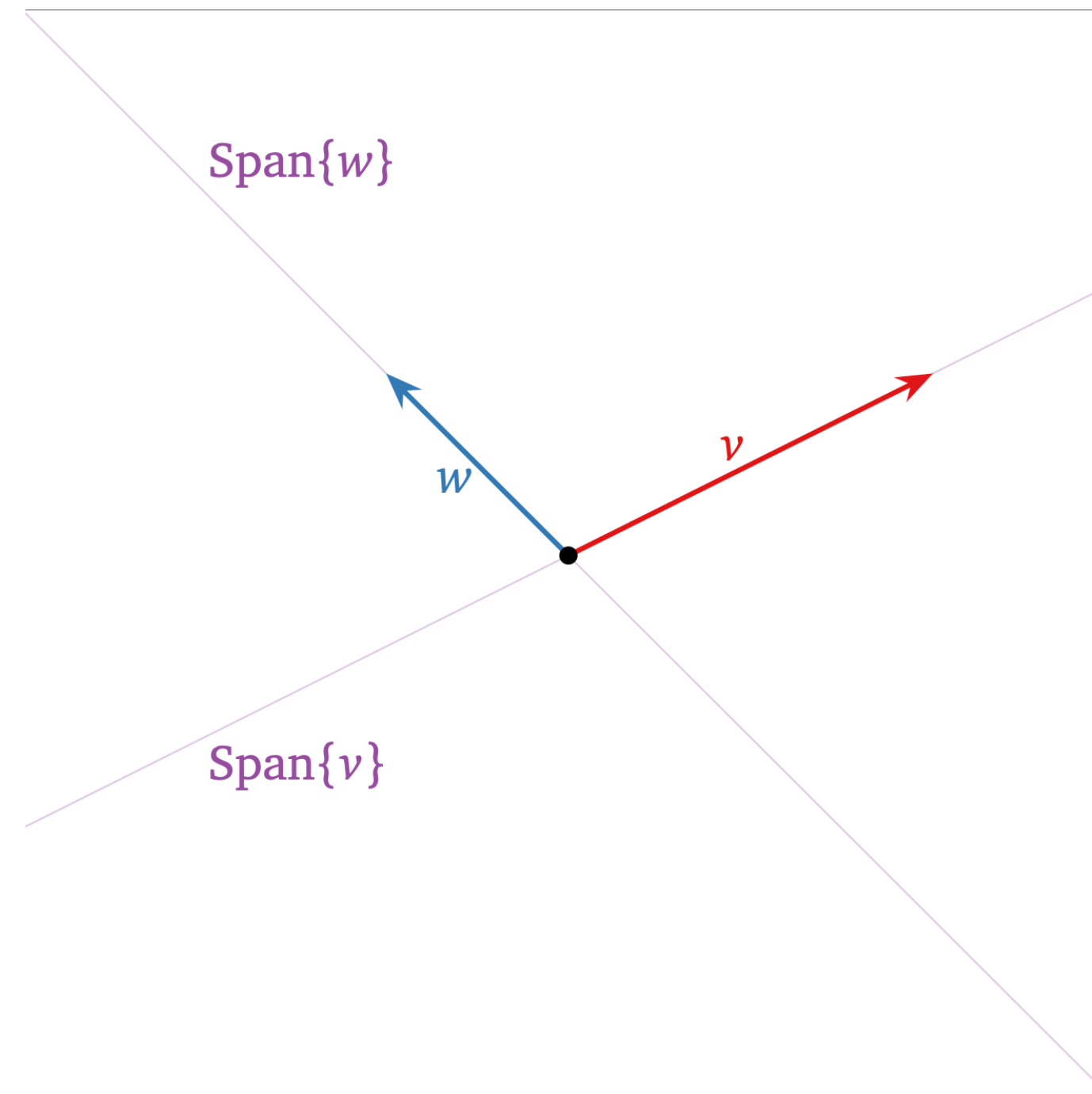


span of 2 vector
a plane

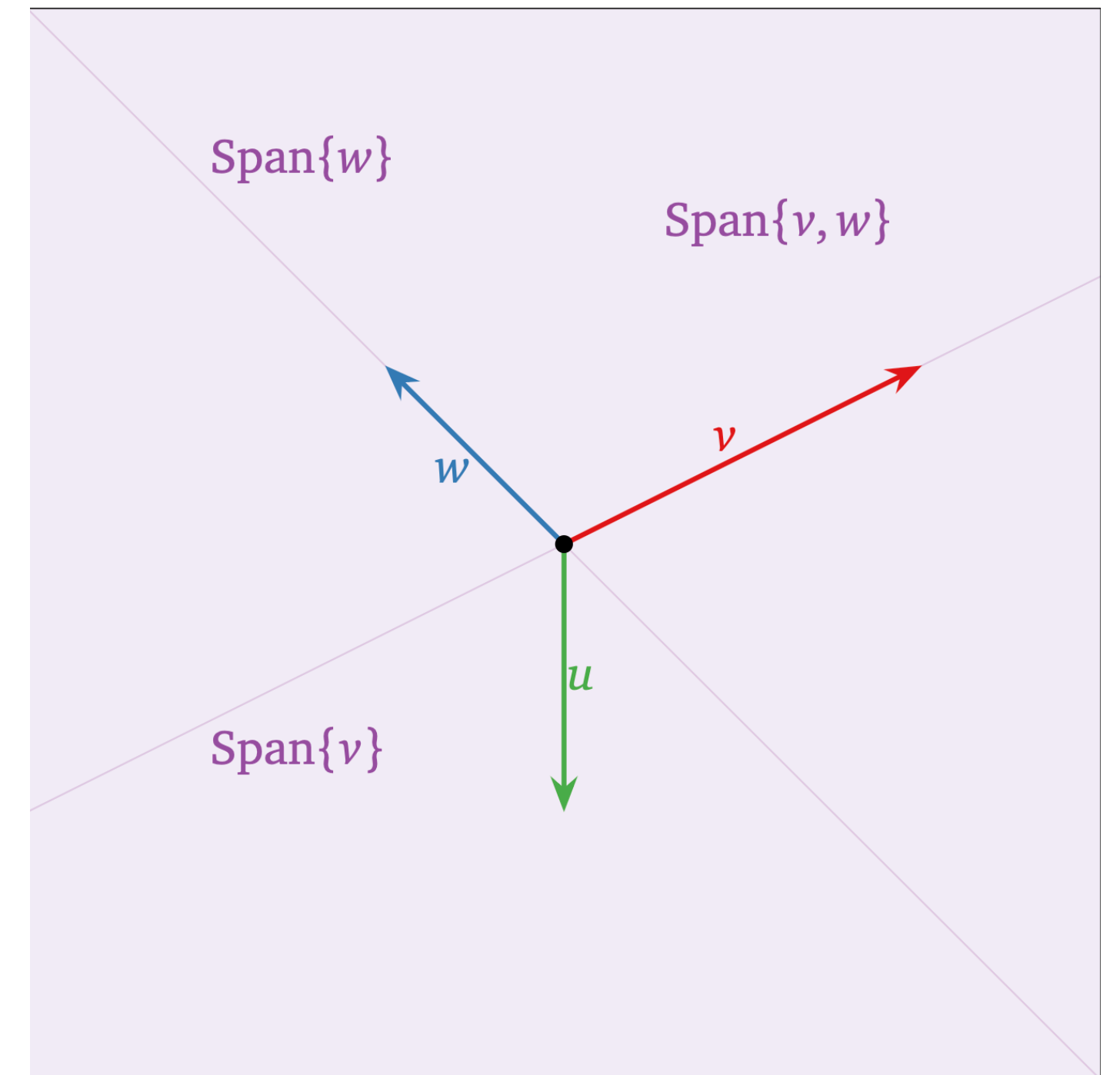
Recap: Increasing Span



span of 1 vector
a line



span of 2 vector
a plane



span of 3 vector
still a plane

Recap: Pivots and Linear Dependence

Theorem. The columns of a matrix A are linearly independent if and only if A has a pivot in every column

Free variables allow for infinitely many
(nontrivial) solution

Recap: Example

$$\mathbf{v}_1 = \begin{bmatrix} -4 \\ 4 \\ 2 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} -3 \\ 6 \\ -3 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} -5 \\ 8 \\ -2 \end{bmatrix}$$

The reduced echelon form of $[\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3]$ is

$$\begin{bmatrix} 1 & 0 & 0.5 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

column
without a
pivot

Recap: Linear Independence and Full Span

The columns of a $(m \times n)$ matrix span all of $\mathbb{R}^n$ if there is a pivot in every row

The columns of a matrix are linearly independent if there is a pivot in every column

Don't confuse these!

Matrix Transformations

Recall: Spans (with Matrices)

Definition. The *span* of a set of vectors is the set of all possible linear combinations

$$\text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\} = \{[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n] \mathbf{v} : \mathbf{v} \in \mathbb{R}^n\}$$

$v_1 \vec{a}_1 + v_2 \vec{a}_2 + \dots + v_n \vec{a}_n$
lin. comb.

$$\mathbf{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$$

Matrices as Transformations

Matrices allow us to *transform* vectors into vectors the span of its columns

$$\mathbf{x} \mapsto A\mathbf{x}$$

map a vector $\mathbf{v}$ to the vector $A\mathbf{v}$

$$\begin{array}{c} \textcolor{red}{A} \\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \end{array}$$

$\vec{x} \mapsto A \vec{x}$
 $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \mapsto \begin{bmatrix} 1 \\ 2 \end{bmatrix}$
 $\begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} \mapsto \begin{bmatrix} 2 \\ 6 \end{bmatrix}$

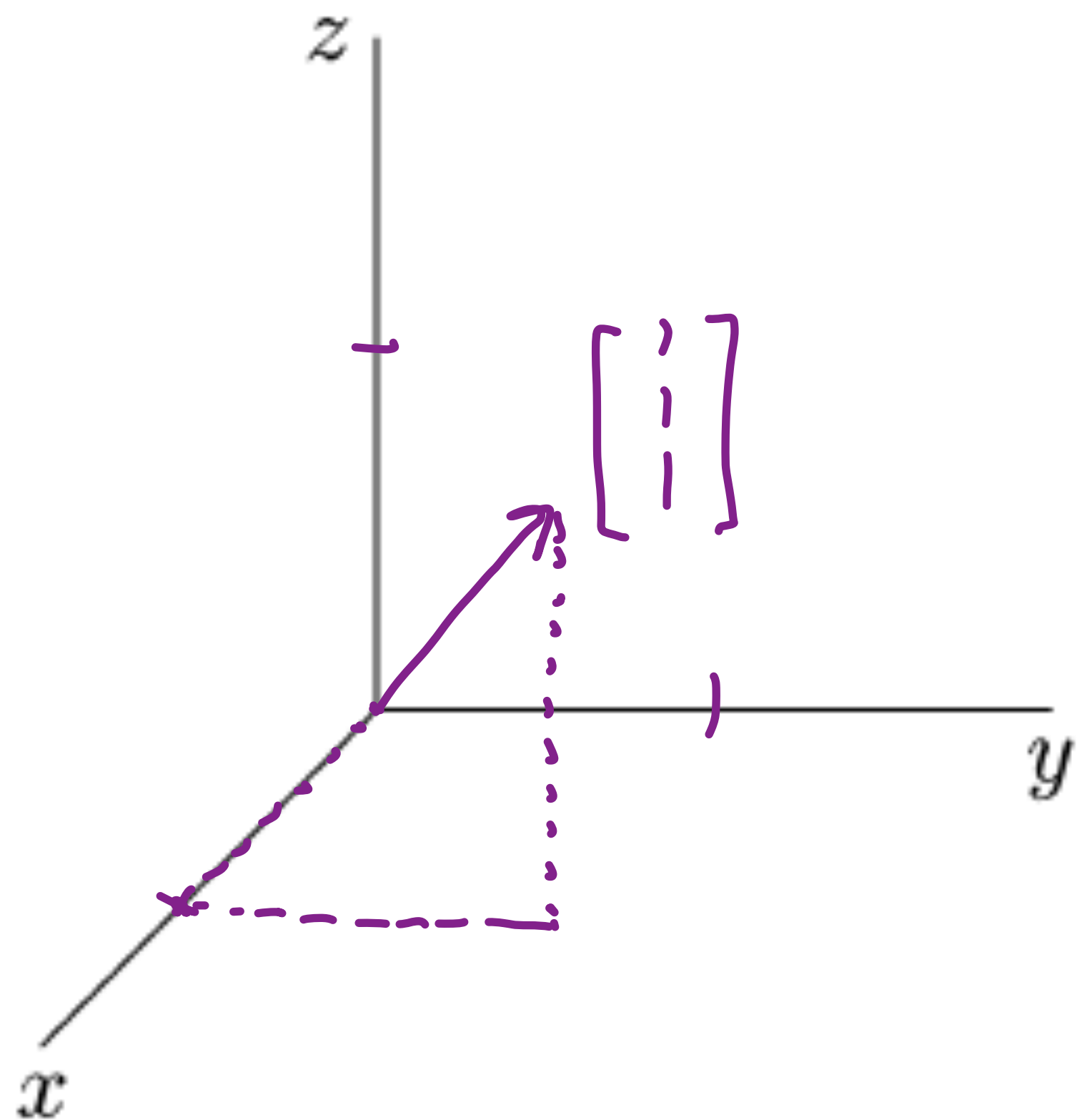
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto \begin{bmatrix} x_1 \\ 2x_2 \end{bmatrix}$$

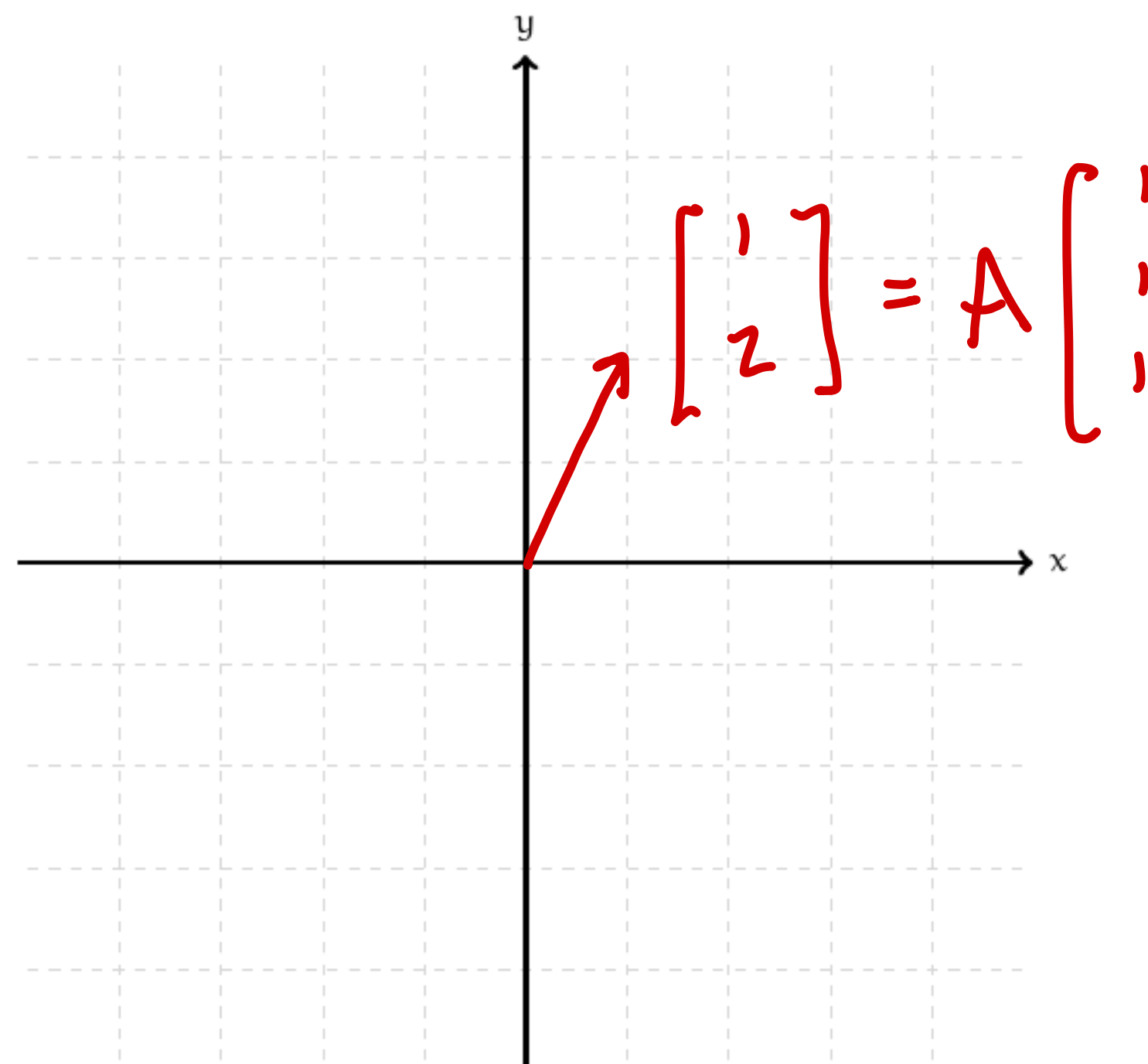
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 2 \end{bmatrix} \rightarrow x_3 \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} x_1 \\ 2x_2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ 2x_2 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} \mathbf{x}$$



!! Important !!

The vector may be a different
size after translation

Recall: Matrix-Vector Multiplication and Dimension

matrix-vector multiplication only works if the number of *columns* of the matrix matches the dimension of the vector

$$\begin{array}{c} \textcolor{blue}{m} \left[\begin{array}{ccc} * & \dots & * \\ * & \dots & * \\ \vdots & \ddots & \vdots \\ * & \dots & * \\ * & \dots & * \end{array} \right] \quad \textcolor{red}{n} \left[\begin{array}{c} * \\ \vdots \\ * \end{array} \right] = \textcolor{blue}{m} \left[\begin{array}{c} * \\ * \\ \vdots \\ * \\ * \end{array} \right] \\ (m \times n) \quad \mathbb{R}^n \quad \mathbb{R}^m \end{array}$$

Motivating Questions

What kind of functions can we define this way?

How do we interpret what the transformation does to a set of vectors?

How does this relate to matrix equations?

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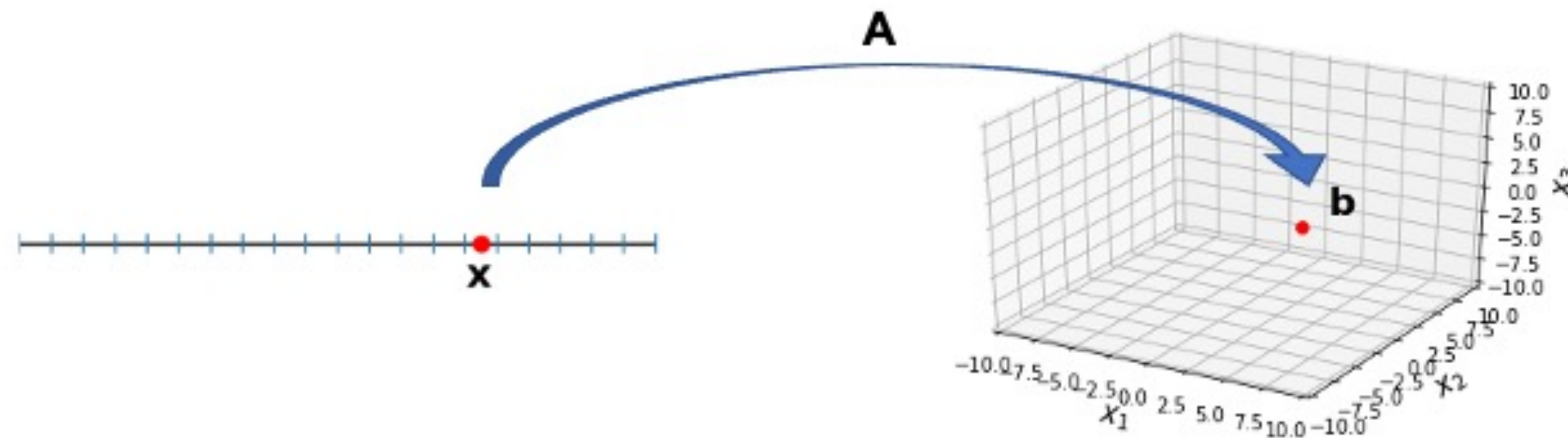
A New Interpretation of the Matrix Equation

$A\mathbf{x} = \mathbf{b}?$ $\equiv$ is there a vector which A
transforms into $\mathbf{b}$?

Solve $A\mathbf{x} = \mathbf{b}$ $\equiv$ find a vector which A
transforms into $\mathbf{b}$

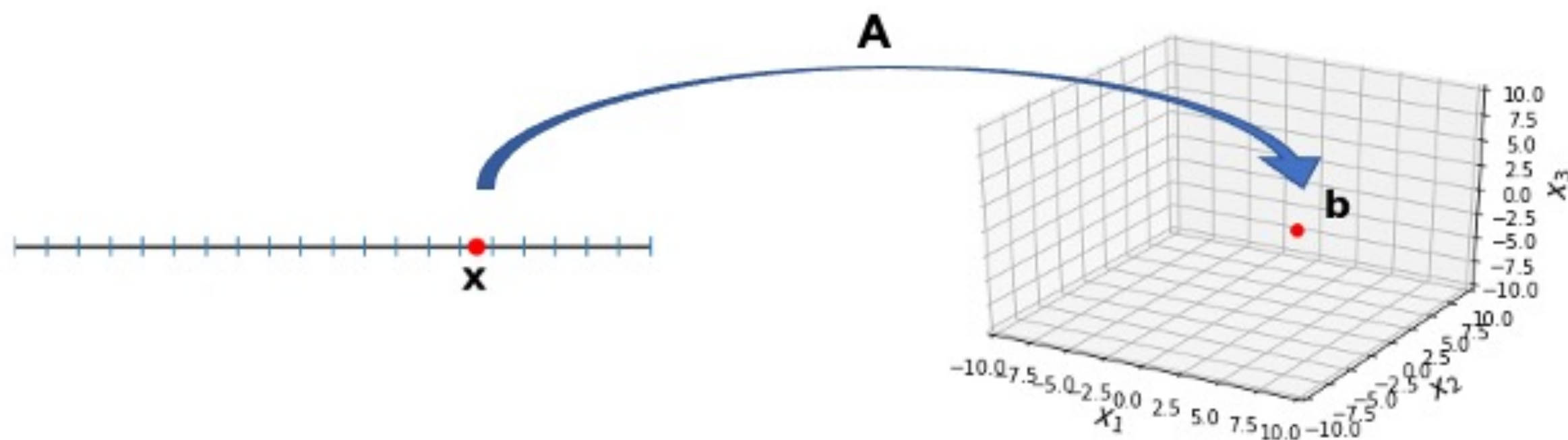
Question

Suppose a matrix transforms a vector according to the following picture. What is the size of the matrix?



$\mathbb{R}^1$

$\mathbb{R}^3$



$\vec{x}$



$$A \vec{x}$$

$$3 \times 2$$

$$3 \times 1$$

$$A$$

$m \times n$

$$3 \times 1$$

$$\mathbb{R}^n = A \mathbb{R}^m$$

$\mathbb{R}^1$ $\mathbb{R}^3$

$$\mathbb{R}^n \rightarrow \mathbb{R}^n$$

Mapping between the same space can be viewed
as a way of moving around points

Transformations

Transformations in General

Definition. A *transformation* T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a function which maps every vector $\mathbf{v}$ in $\mathbb{R}^n$ to a vector $T(\mathbf{v})$ in $\mathbb{R}^m$

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$$T : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

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$$T : \underbrace{\mathbb{R}^n}_{\text{domain}} \rightarrow \underbrace{\mathbb{R}^m}_{\text{codomain}}$$

Image and Range

Image and Range

Definition. For a vector $\mathbf{v}$, the *image* of $\mathbf{v}$ under the transformation T is the vector $T(\mathbf{v})$

$$V \subseteq \mathbb{R}^n = \text{dom}(T) \quad \text{image}_T(V) = \{T(\vec{v}) : \vec{v} \in V\}$$

Image and Range

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Definition. The *range* of a transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is the set of all possible images under T

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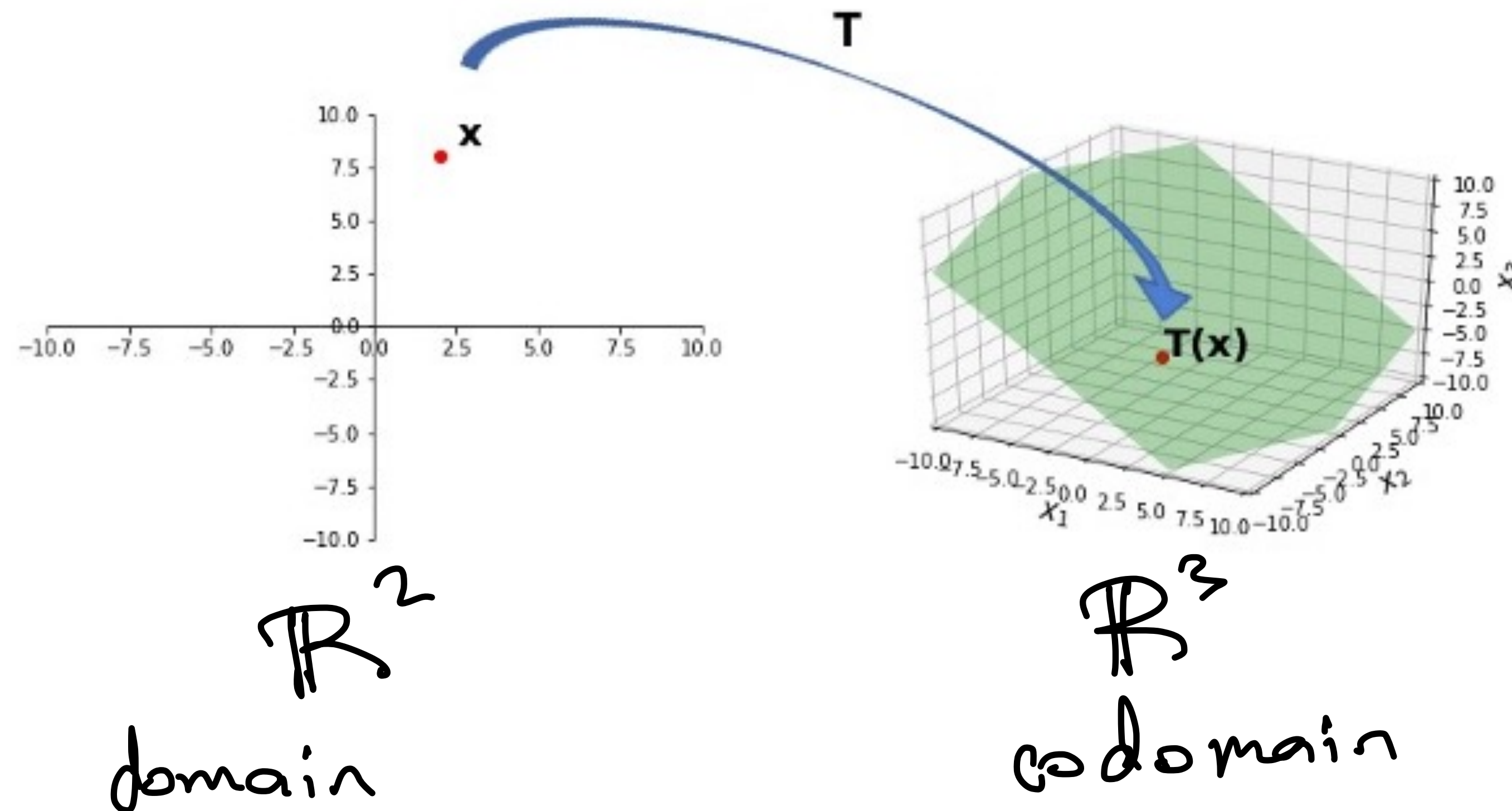
$$\text{ran}(T) = \{T(\mathbf{v}) : \mathbf{v} \in \mathbb{R}^n\}$$

image of $\mathbf{v}$ under $T \equiv$ output of T applied to $\mathbf{v}$

range of $T \equiv$ all possible output of T

Codomain and Range

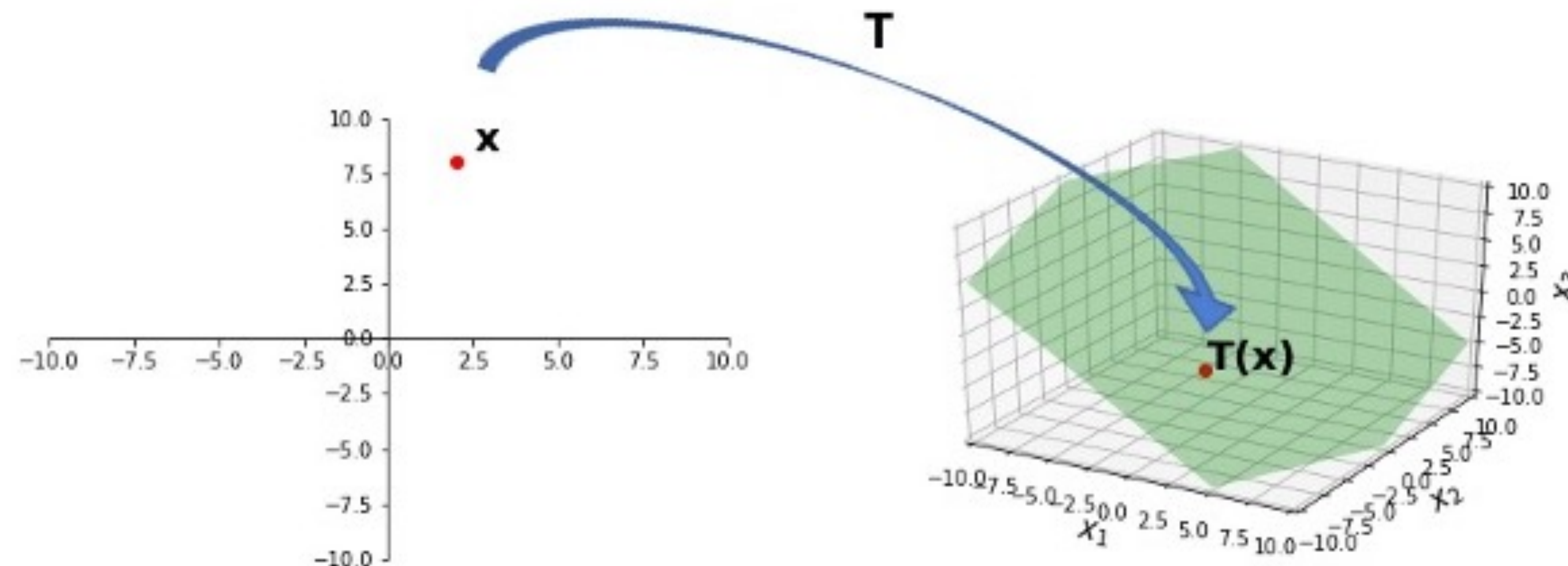
The codomain and range of a transformation may or may not be the same



$$\text{ran}(T) \neq \mathbb{R}^3$$

Codomain and Range

The codomain and range of a transformation may or may not be the same



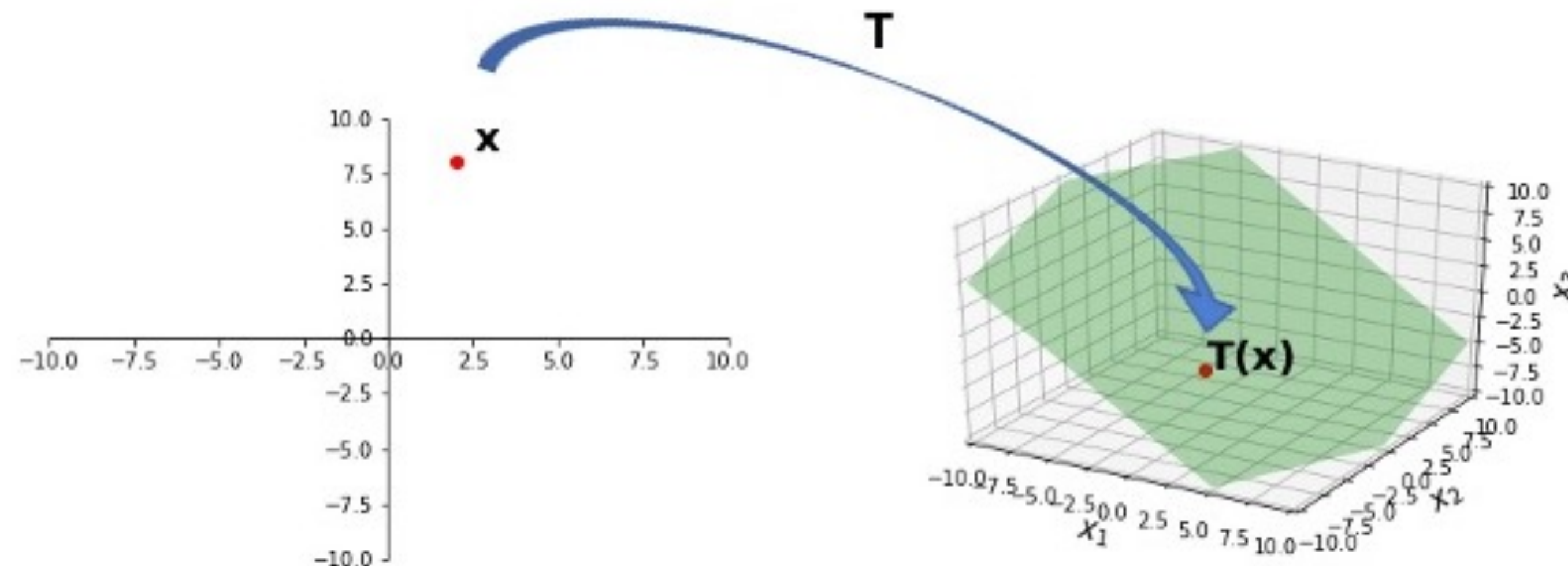
domain: $\mathbb{R}^2$

codomain: $\mathbb{R}^3$

range: just
the green
plane

Codomain and Range

The codomain and range of a transformation may or may not be the same



domain: $\mathbb{R}^2$

codomain: $\mathbb{R}^3$

range: just
the green
plane

The range is always contained in the codomain

Example

This is
not matrix
transformation

$$T: \underset{\text{dom}}{\mathbb{R}^2} \rightarrow \underset{\text{cod}}{\mathbb{R}^3}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mapsto \begin{bmatrix} x_1^2 \\ x_2 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \notin \text{ran}(T)$$

$$\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \in \mathbb{R}^3 = \text{cod}(T)$$

$$\text{ran}(T) = \left\{ \begin{bmatrix} v_1 \\ v_2 \\ 0 \end{bmatrix} : v_1 \geq 0 \right\}$$

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \mapsto \begin{bmatrix} v_1 \\ v_2 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 2 \\ -7 \end{bmatrix} \mapsto \begin{bmatrix} 4 \\ -7 \\ 0 \end{bmatrix}$$

Matrix Transformations

Transformation of a Matrix

Transformation of a Matrix

The *transformation of a* $(m \times n)$ *matrix* A is the function $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that

$$T(\mathbf{v}) = A\mathbf{v}$$

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given $\mathbf{v}$, return A multiplied by $\mathbf{v}$

Transformation of a Matrix

The ***transformation of a*** $(m \times n)$ ***matrix*** A is the function $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that

$$T(\mathbf{v}) = A\mathbf{v}$$

given $\mathbf{v}$, return A multiplied by $\mathbf{v}$

e.g. $T(\mathbf{v}) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \mathbf{v}$

$$T\left(\begin{bmatrix} 3 \\ 5 \end{bmatrix}\right) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 8 \\ 5 \end{bmatrix}$$

Range and Span

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The span of the columns of a matrix A is the set of all possible *images* under A

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$$\text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\} = \text{ran}([\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n])$$

Range and Span

The span of the columns of a matrix A is the set of all possible *images* under A

$$\text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\} = \text{ran}([\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n])$$

The transformation of a vector $\mathbf{v}$ under the matrix A always lies in the span of its columns

Motivating Questions

What kind of functions can we define this way?

How do we interpret what the transformation does to a set of vectors?

How does this relate to matrix equations?

Linear Transformations

Recall: Algebraic Properties

Matrix-vector multiplication satisfies the following two properties:

1. $A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}$ (additivity)

2. $A(c\mathbf{v}) = c(A\mathbf{v})$ (homogeneity)

Example

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \left(\begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \boxed{\begin{bmatrix} 7 \\ 4 \end{bmatrix}}$$

//

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix} + - \boxed{\begin{bmatrix} 7 \\ 4 \end{bmatrix}}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Example

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \left(2 \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 6 \end{bmatrix} = \boxed{\begin{bmatrix} 10 \\ 6 \end{bmatrix}}$$
$$2 \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \boxed{\begin{bmatrix} 10 \\ 6 \end{bmatrix}}$$

Linear Transformations

Definition. A transformation $T: \mathbb{R}^m \rightarrow \mathbb{R}^n$ is *linear* if it satisfies the following two properties

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ (additivity)

2. $T(c\mathbf{v}) = cT(\mathbf{v})$ (homogeneity)

Linear Transformations

Definition. A transformation $T: \mathbb{R}^m \rightarrow \mathbb{R}^n$ is *linear* if it satisfies the following two properties

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ (additivity)

2. $T(c\mathbf{v}) = cT(\mathbf{v})$ (homogeneity)

Matrix transformations are linear transformations

Example: Identity

$$T(\mathbf{v}) = \mathbf{v}$$

(add) $T(\vec{u} + \vec{v}) = \vec{u} + \vec{v}$ ✓

$$T(\vec{u}) + T(\vec{v}) = \vec{u} + \vec{v}$$

(homogen.) $T(c \vec{u}) = c \vec{u}$ ✓

$$c T(\vec{u}) = c \vec{u}$$

Example: Zero

$$T(\mathbf{v}) = \mathbf{0}$$

(add) $T(\vec{u} + \vec{v}) = \vec{0} \quad //$

$$T(\vec{u}) + T(\vec{v}) = \vec{0} + \vec{0} = \vec{0} \quad \checkmark$$

(homogen) $T(c\vec{u}) = \vec{0} \quad //$

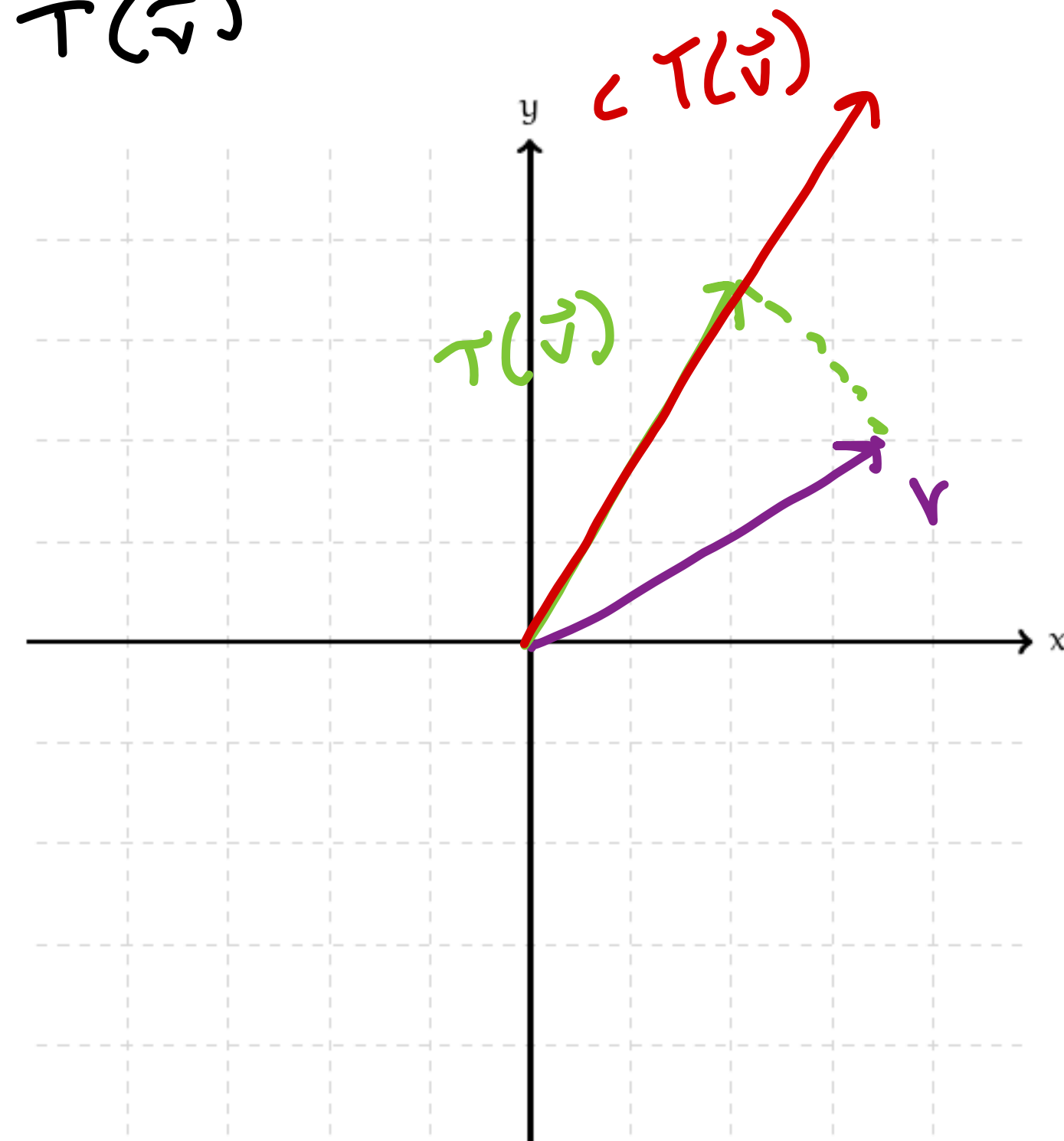
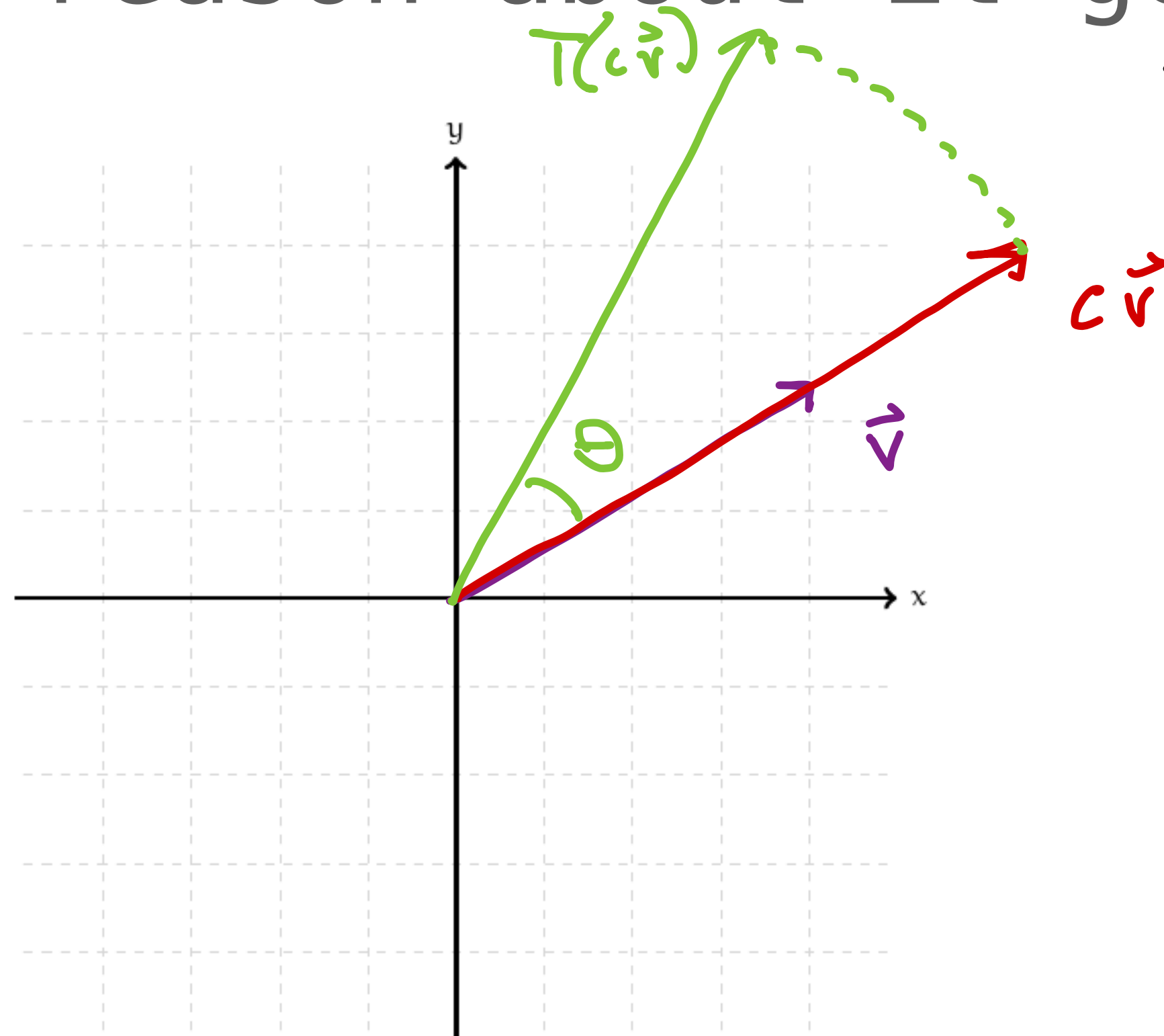
$$c T(\vec{u}) = c \vec{0} = \vec{0} \quad \checkmark$$

Example: Rotation

We'll see this on Thursday, but we can reason about it geometrically for now.

homogen.

$$T(c\vec{v}) = c T(\vec{v})$$



Example: Indefinite Integrals

$$T(f) = \int f(x) dx$$

Disclaimer:
Advanced
Material

the same goes for derivatives
(how are functions vectors???)

Example: Expectation

$$T(X) = \mathbb{E}[X]$$

Disclaimer:
Advanced
Material

This is exactly linearity of expectation.

(how are random variables vectors???)

Non-Example: Squares

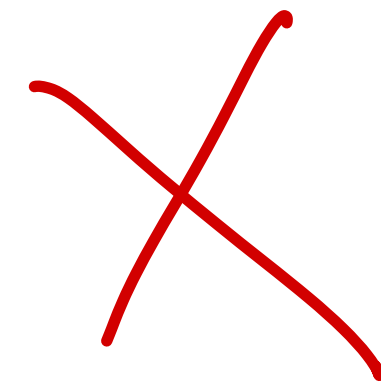
$$T(x) = x^2$$

Note that $T: \mathbb{R}^1 \rightarrow \mathbb{R}^1$

(add)

$$T(1+2) = 9 \neq$$

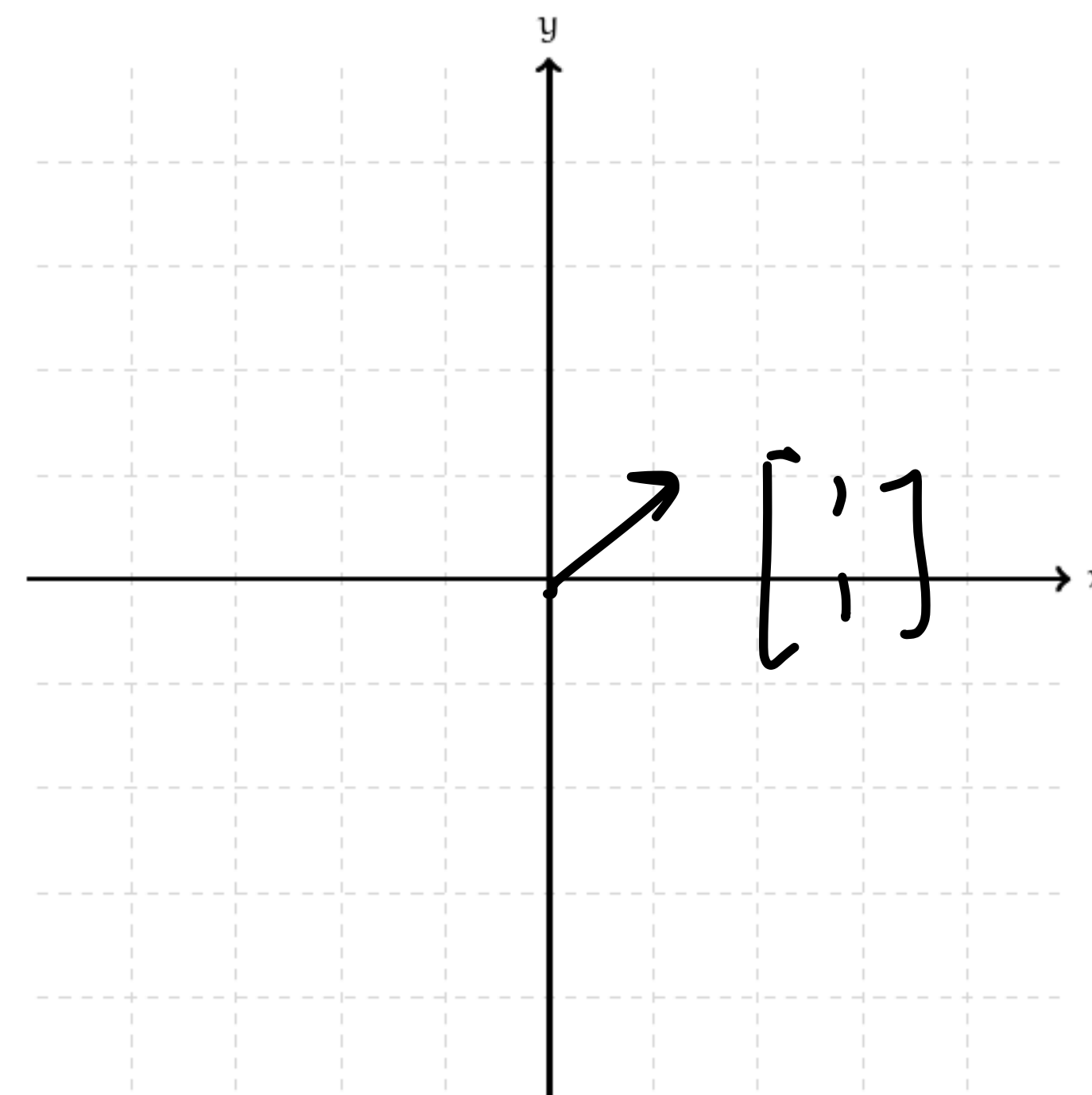
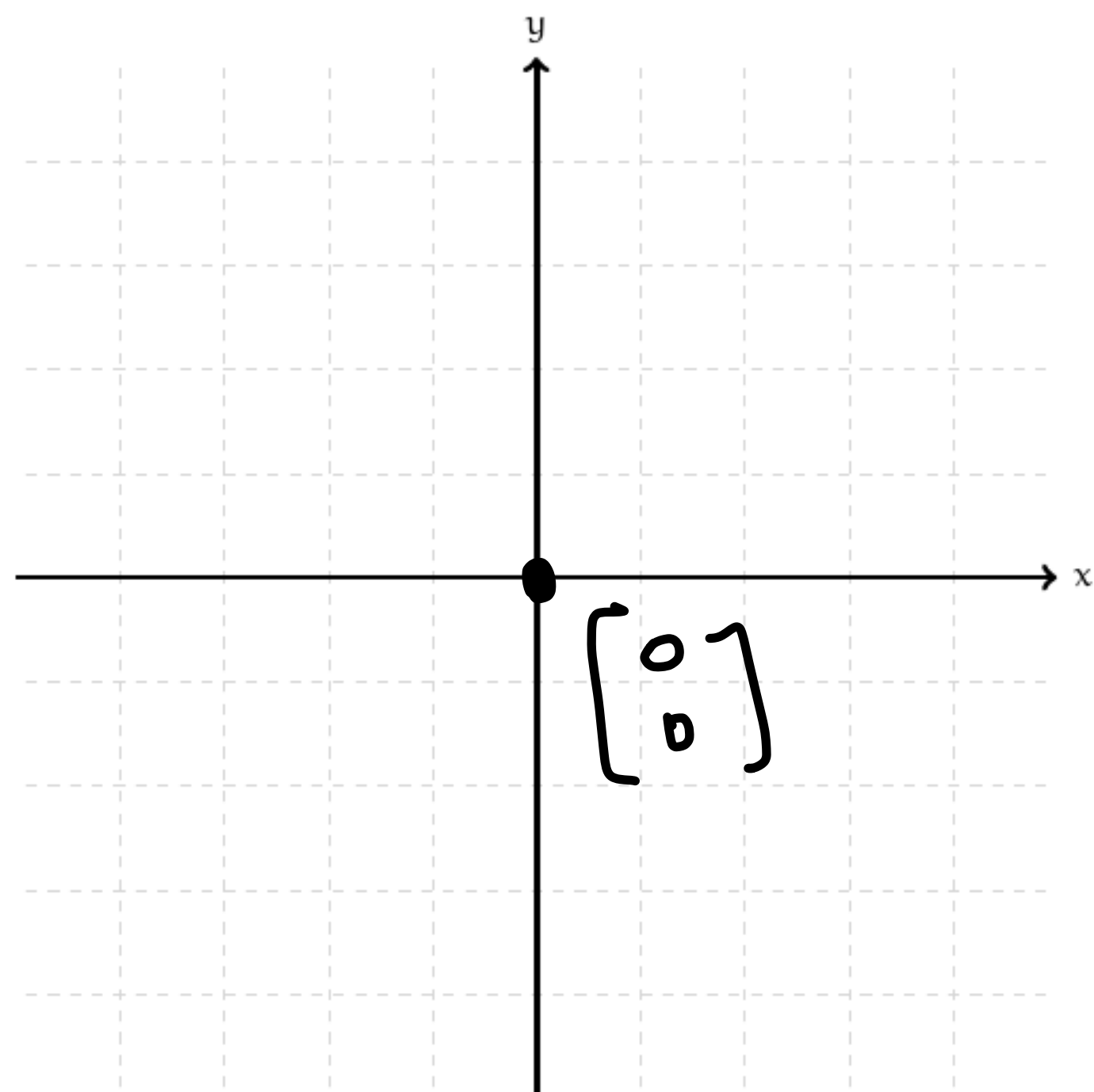
$$T(1) + T(2) = 1 + 4 = 5$$



Non-Example: Translation

if T is lin. $\Rightarrow T(\vec{0}) = T(0 \begin{bmatrix} 1 \\ 1 \end{bmatrix}) = 0 T(\begin{bmatrix} 1 \\ 1 \end{bmatrix}) = \vec{0}$

$$\mathbf{x} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$



Properties of Linear Transformations

The Zero Vector

$$T(\mathbf{0}) = ???$$

The Zero Vector

$$T(\mathbf{0}) = \mathbf{0}$$

The Zero Vector

$$T(\mathbf{0}) = \mathbf{0}$$

The zero vector is *fixed* by linear transformations

The Zero Vector

$$T(\mathbf{0}) = \mathbf{0}$$
Two red arrows originate from the text 'Note: These may be different dimensions!' below. One arrow points diagonally up and to the left towards the bold zero vector inside the parentheses of the equation $T(\mathbf{0}) = \mathbf{0}$. The other arrow points diagonally up and to the right towards the bold zero vector on the right side of the equation.

Note: These may be different dimensions!

The zero vector is *fixed* by linear transformations

A Single Condition

$$T(a\mathbf{v} + b\mathbf{u}) = aT(\mathbf{v}) + bT(\mathbf{u})$$

A Single Condition

$$T(a\mathbf{v} + b\mathbf{u}) = aT(\mathbf{v}) + bT(\mathbf{u})$$

We can combine our linearity conditions

$$\begin{aligned} T(a\vec{v} + b\vec{u}) &= T(a\vec{v}) + T(b\vec{u}) \\ &= aT(\vec{v}) + bT(\vec{u}) \end{aligned}$$

Ex. Show if $p \Rightarrow q$ holds, then T is Lin.

A Single Condition

Theorem. A transformation $T: \mathbb{R}^m \rightarrow \mathbb{R}^n$ is linear if and only if for any vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^m$ and any real numbers a and b ,

$$T(a\mathbf{u} + b\mathbf{v}) = aT(\mathbf{u}) + bT(\mathbf{v})$$

A Single Condition

Theorem. A transformation $T: \mathbb{R}^m \rightarrow \mathbb{R}^n$ is linear if and only if for any vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^m$ and any real numbers a and b ,

$$T(a\mathbf{u} + b\mathbf{v}) = aT(\mathbf{u}) + bT(\mathbf{v})$$

It's often easiest to show this
single condition

Linear Combinations

$$T(a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_n \mathbf{v}_n)$$

And we can generalize to any linear combination

$$T\left(\sum_{i=1}^n a_i \vec{v}_i\right) = \sum_{i=1}^n T(a_i \vec{v}_i) = \sum_{i=1}^n a_i T(\vec{v}_i)$$

Geometry of Matrix Transformations

Motivating Questions

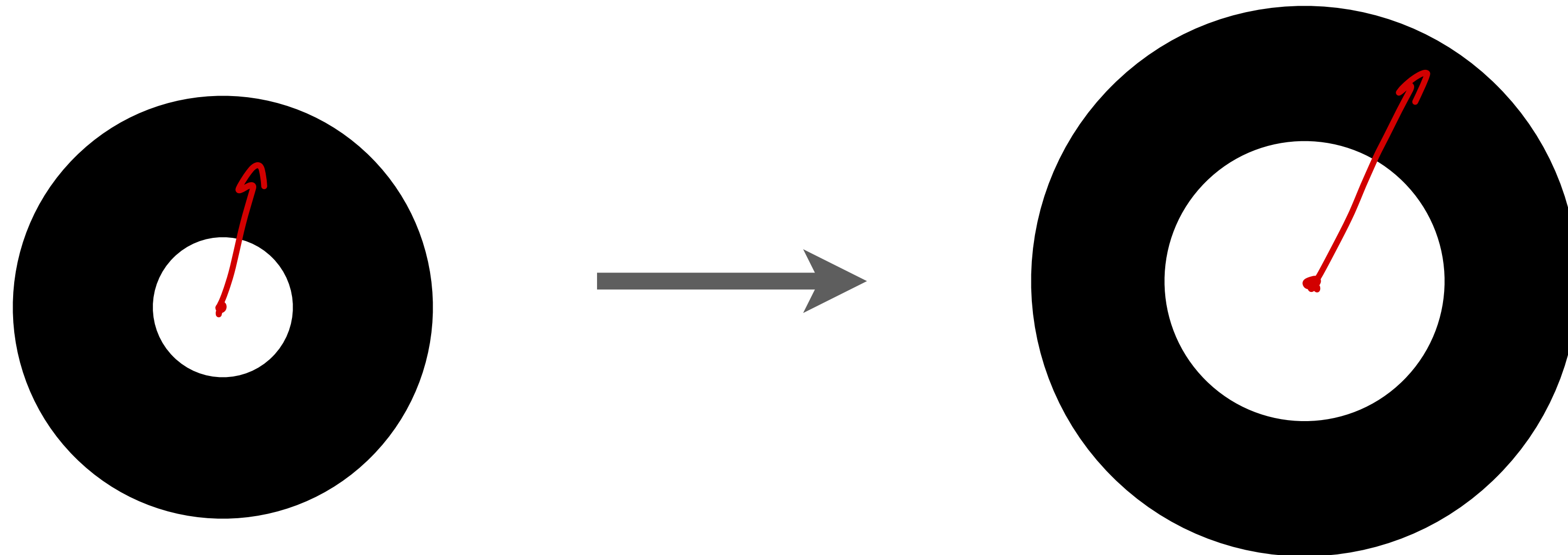
What kind of functions can we define this way?

How do we interpret what the transformation does to a set of vectors?

How does this relate to matrix equations?

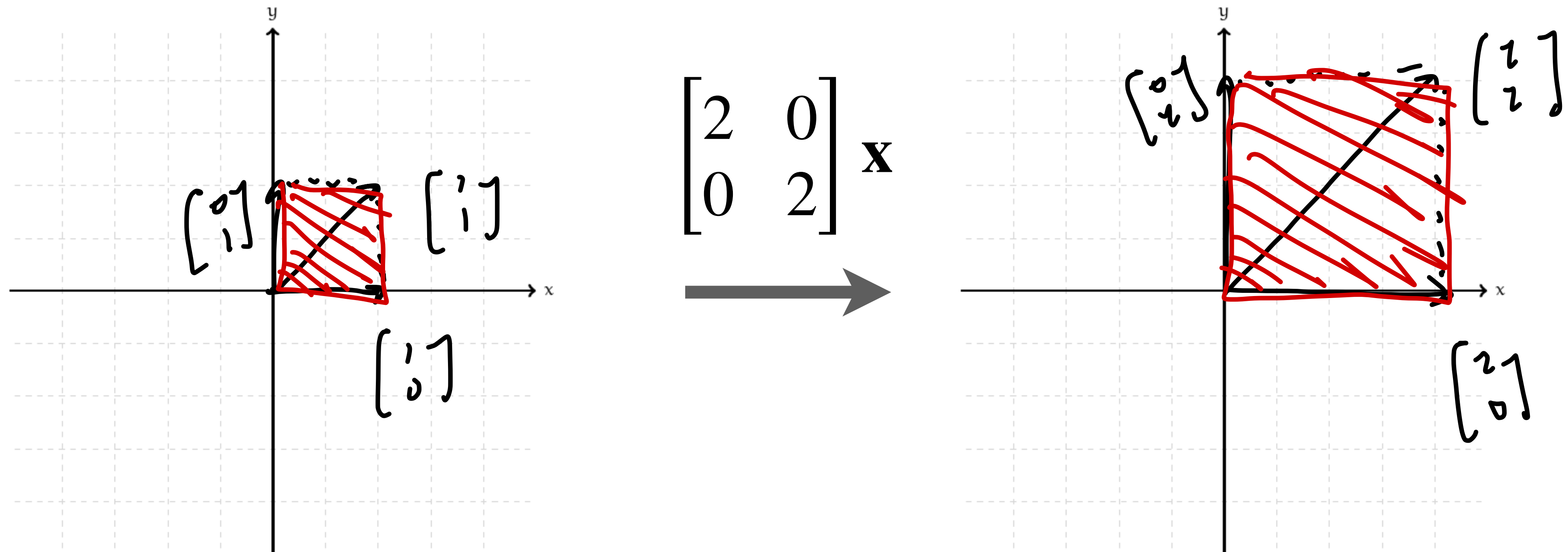
Matrix transformations change the
"shape" of a set of set of
vectors (points)

Example: Dilation



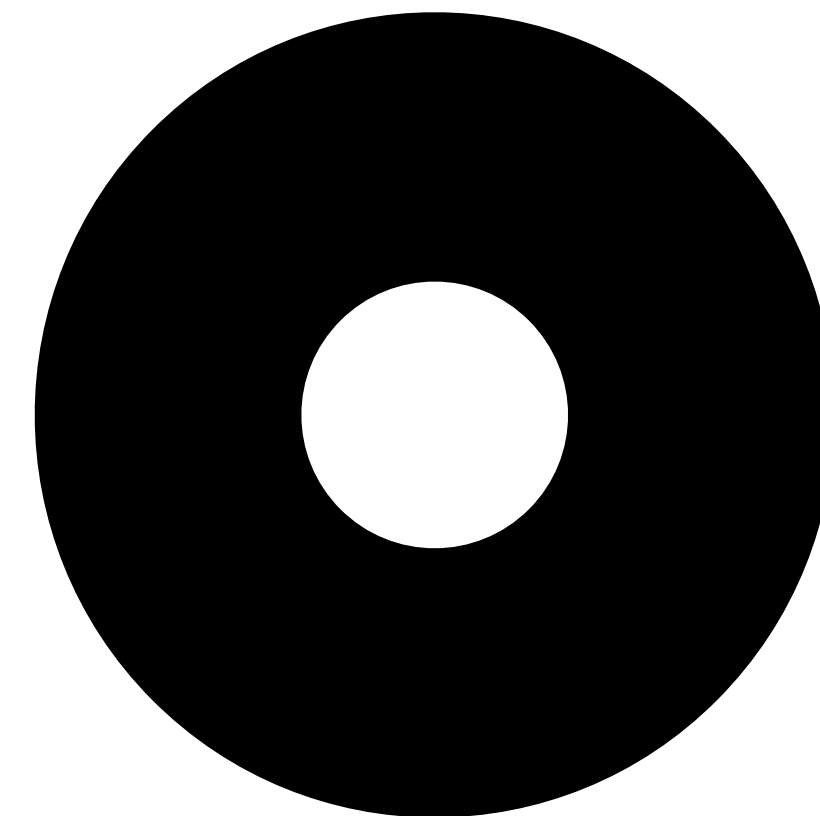
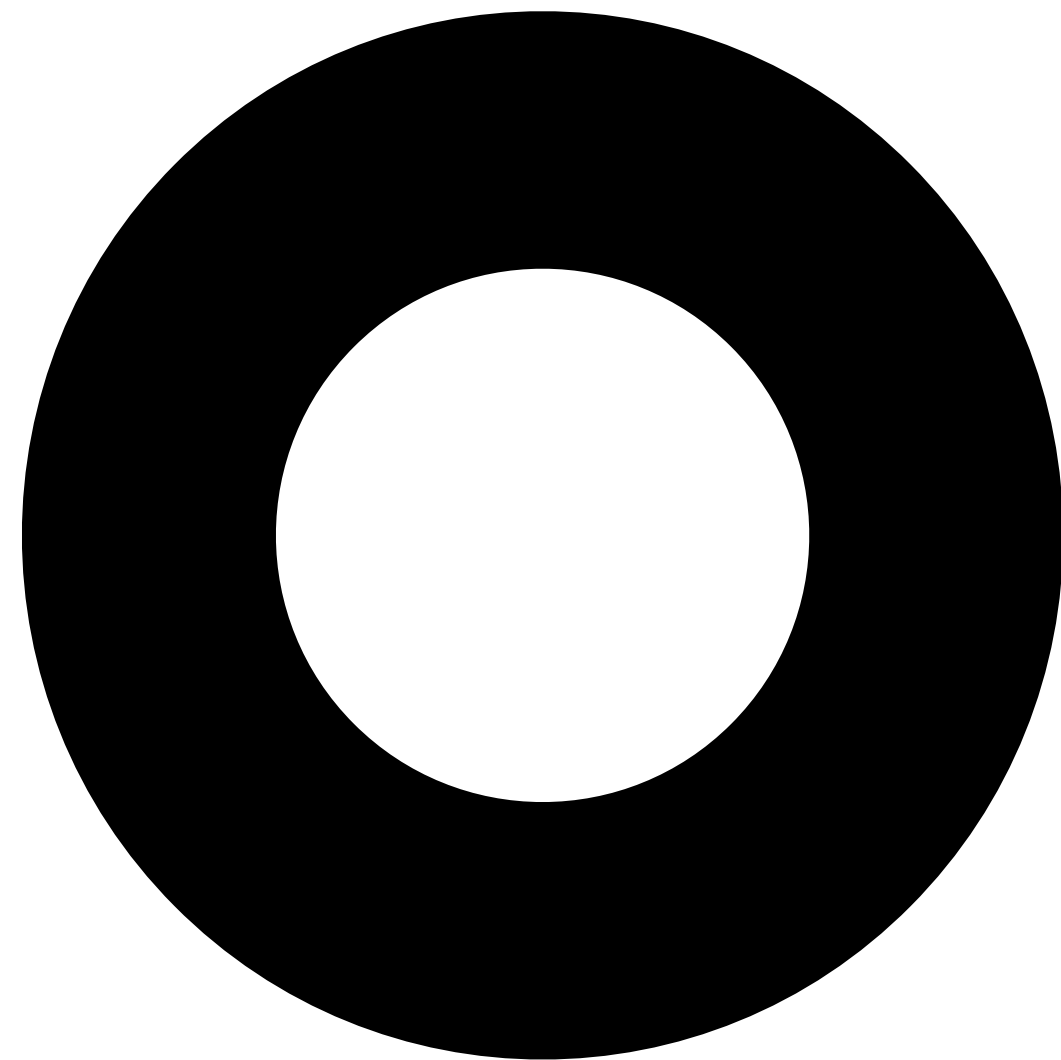
Example: Dilation

$$\begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} rx_1 \\ rx_2 \end{bmatrix}$$



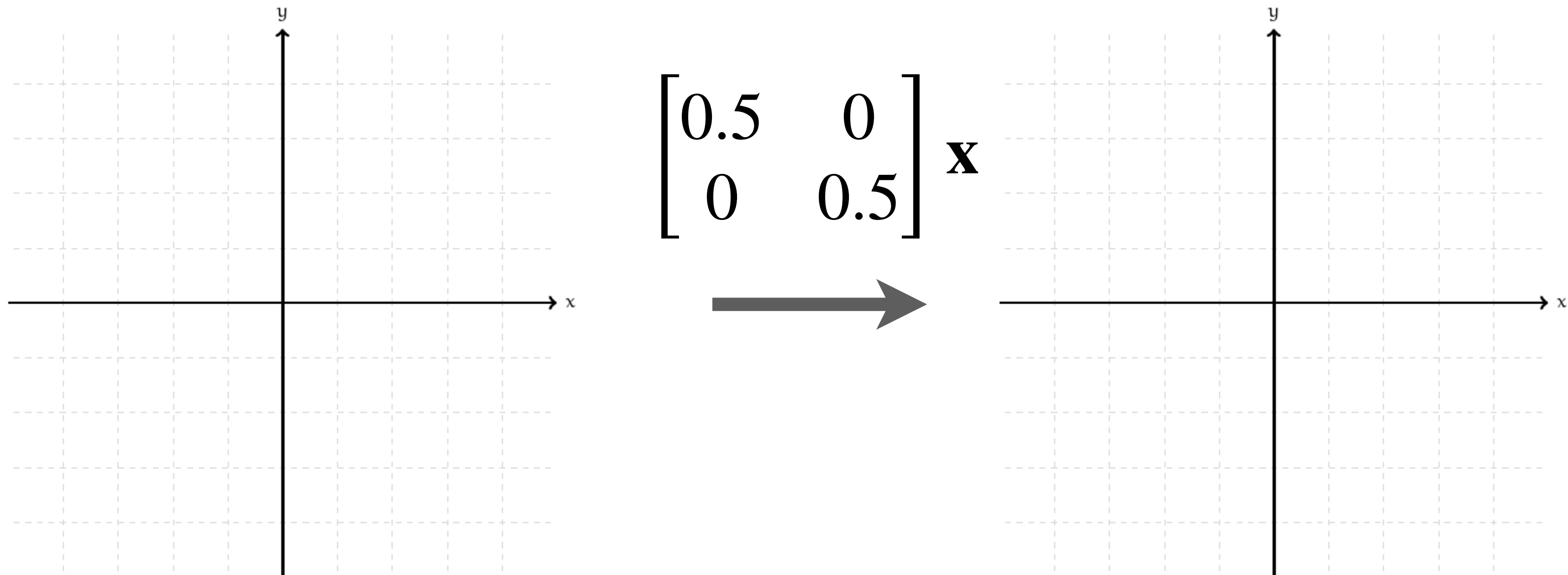
if $r > 1$, then the transformation pushes points away from the origin.

Example: Contraction



Example: Contraction

$$\begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} rx_1 \\ rx_2 \end{bmatrix}$$



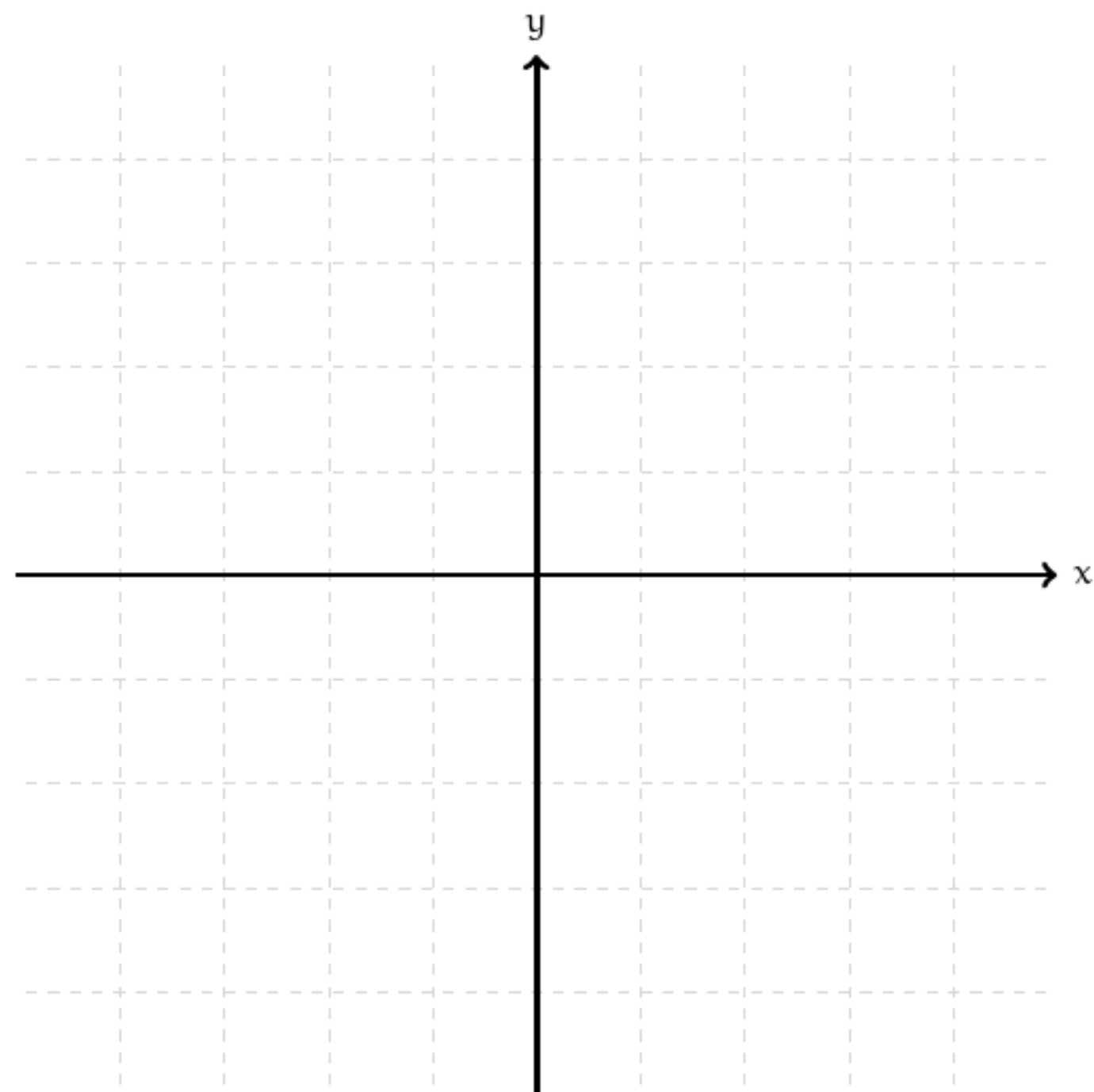
if $0 \leq r \leq 1$, then the transformation
pulls points towards the origin.

Example: Shearing

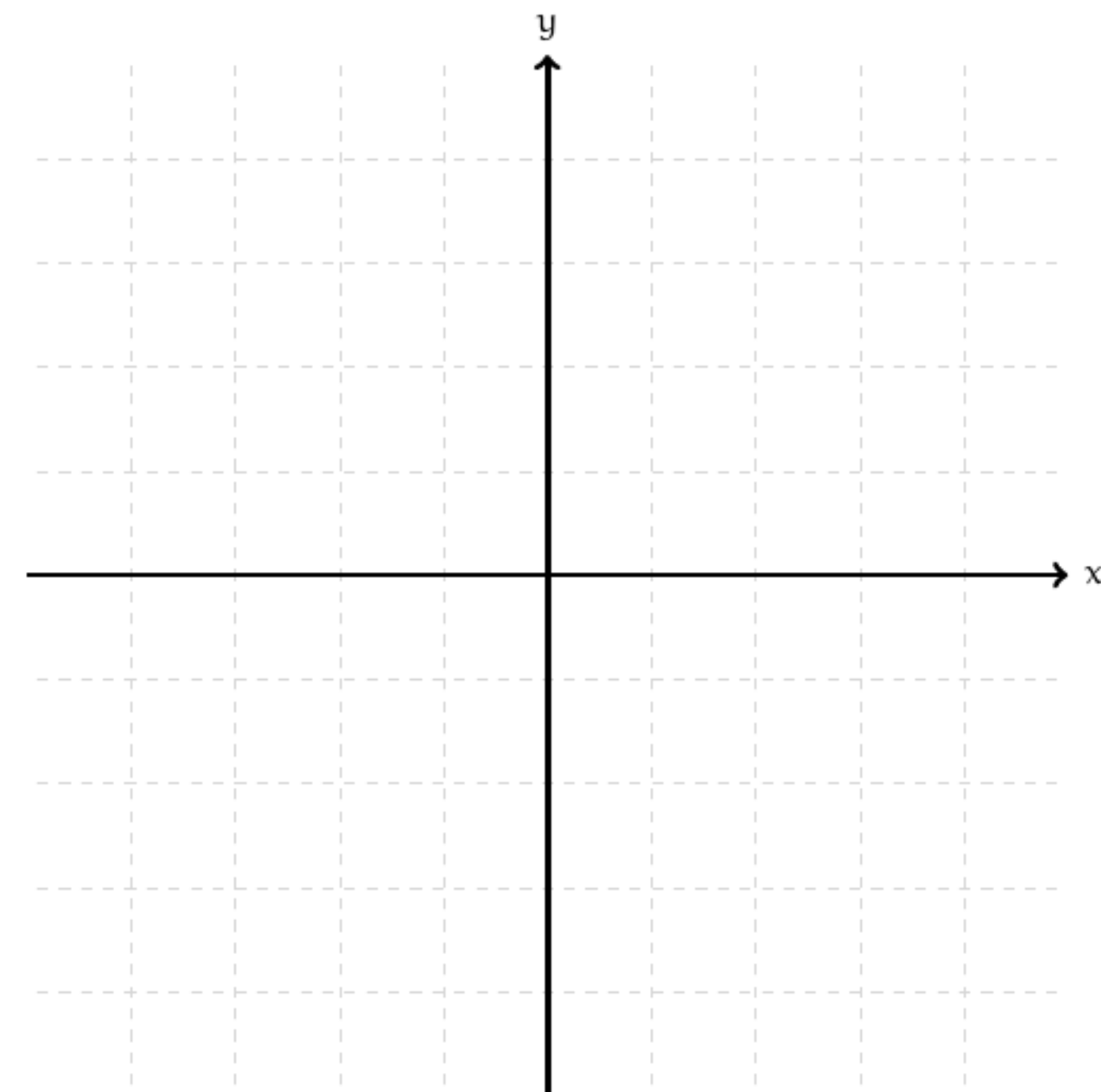


Example: Shearing

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ x_2 \end{bmatrix}$$

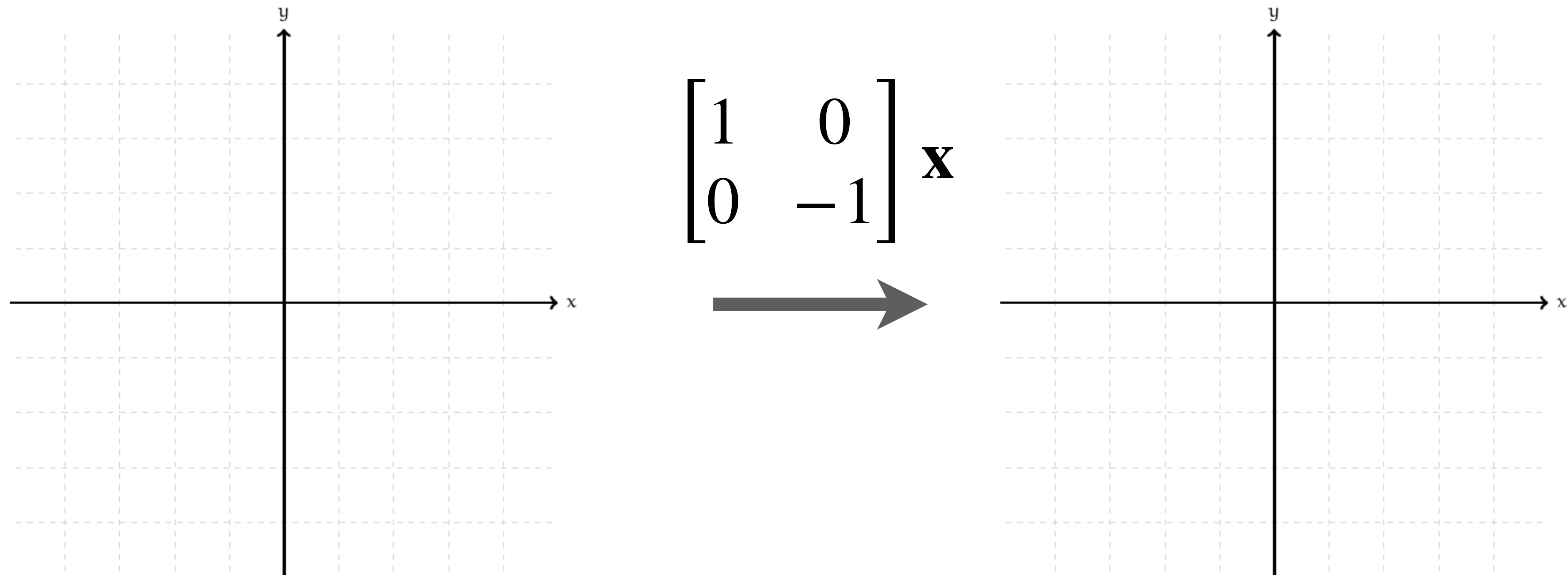


$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \mathbf{x}$$



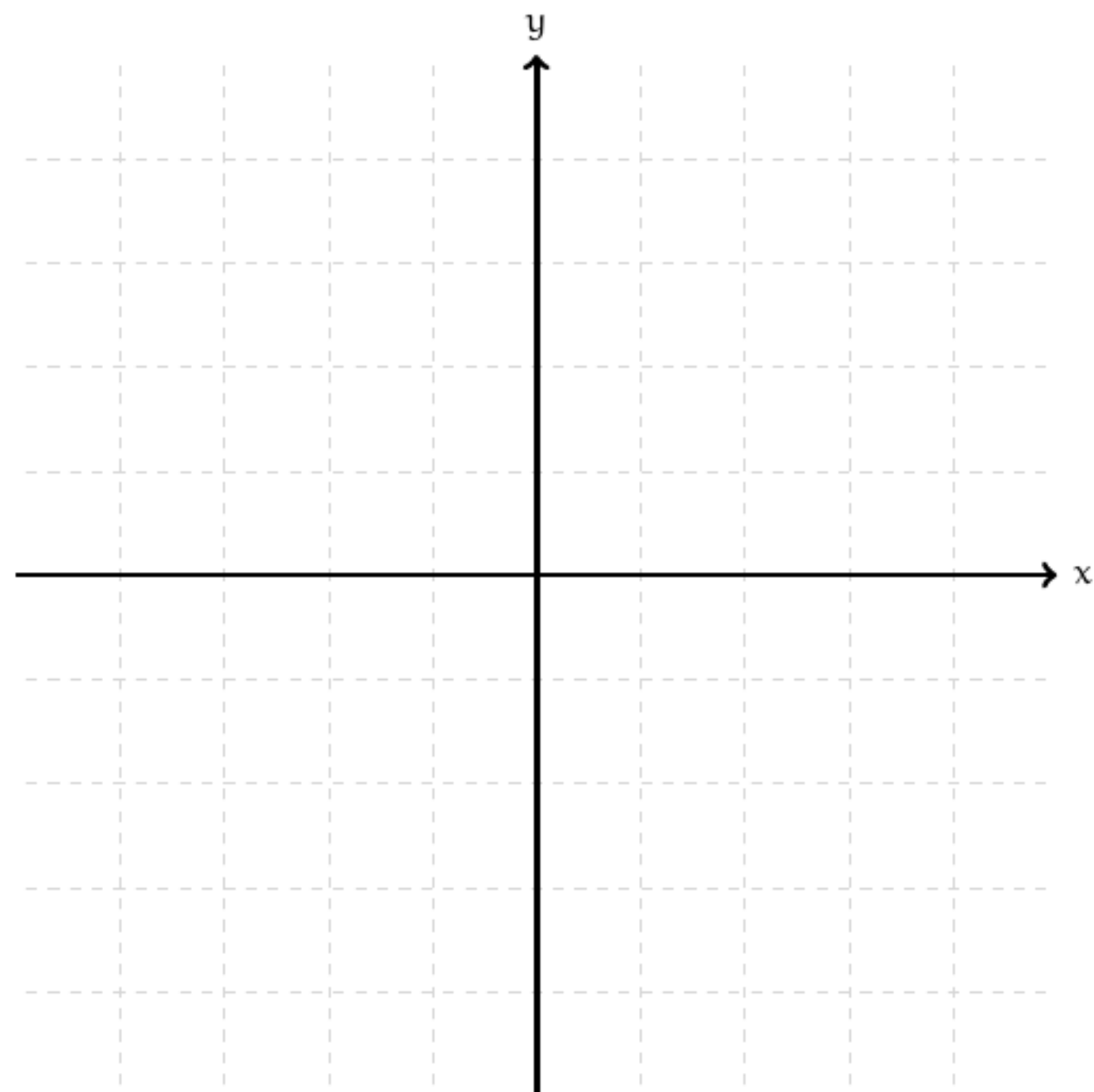
Imagine shearing like with rocks or metal.

Question

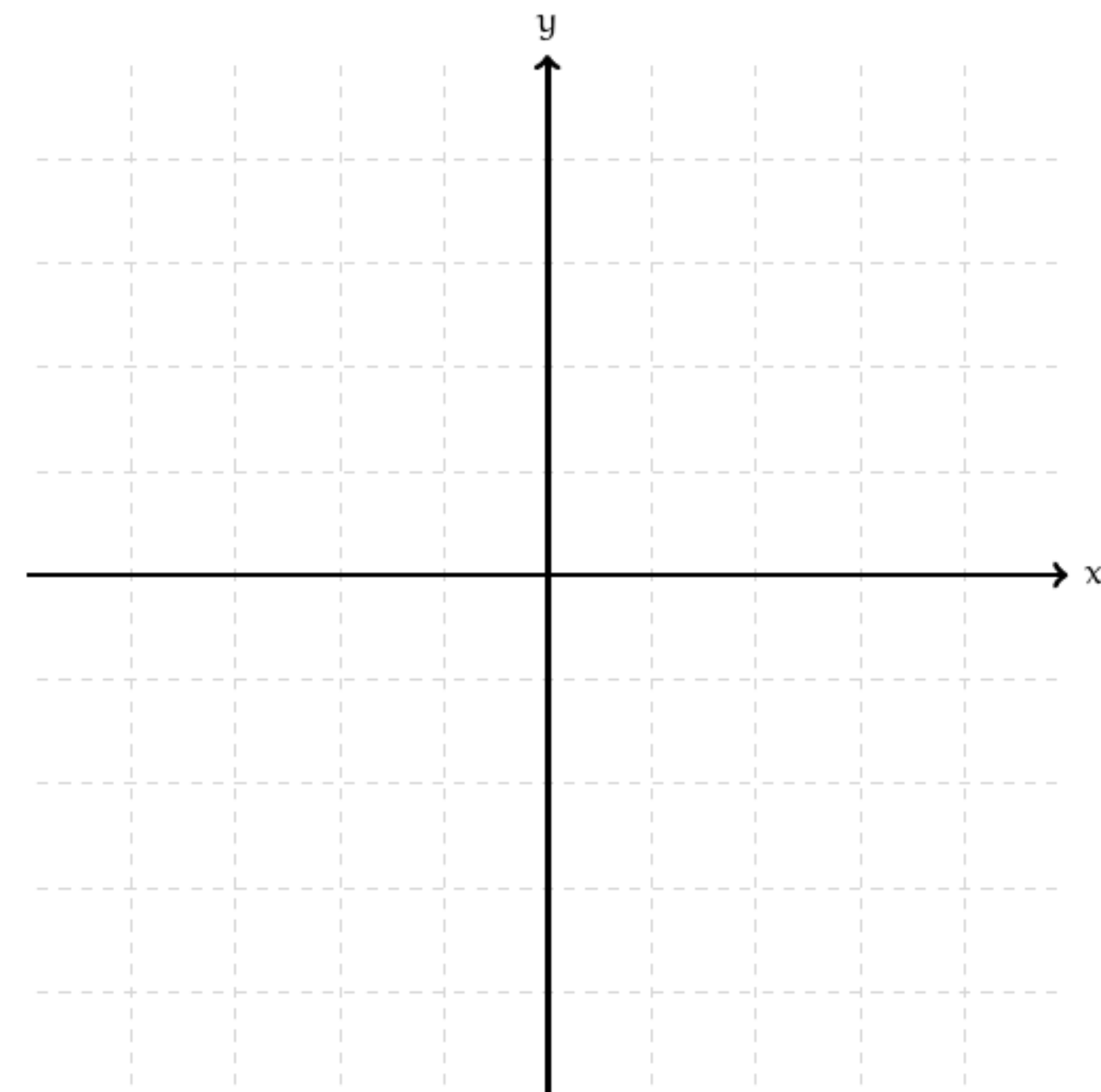


Draw how this matrix transforms points. What kind of transformation does it represent?

Answer: Reflection



$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \mathbf{x}$$



demo

Summary

Matrices can be viewed as **linear transformations**

Matrix transformations change the **shape** of points sets

Linear transformations behave well with respect to **linear combinations**